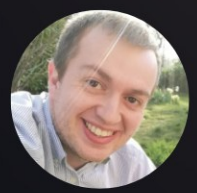
Simulation-based inference for exoplanets



Gilles Louppe
g.louppe@uliege.be



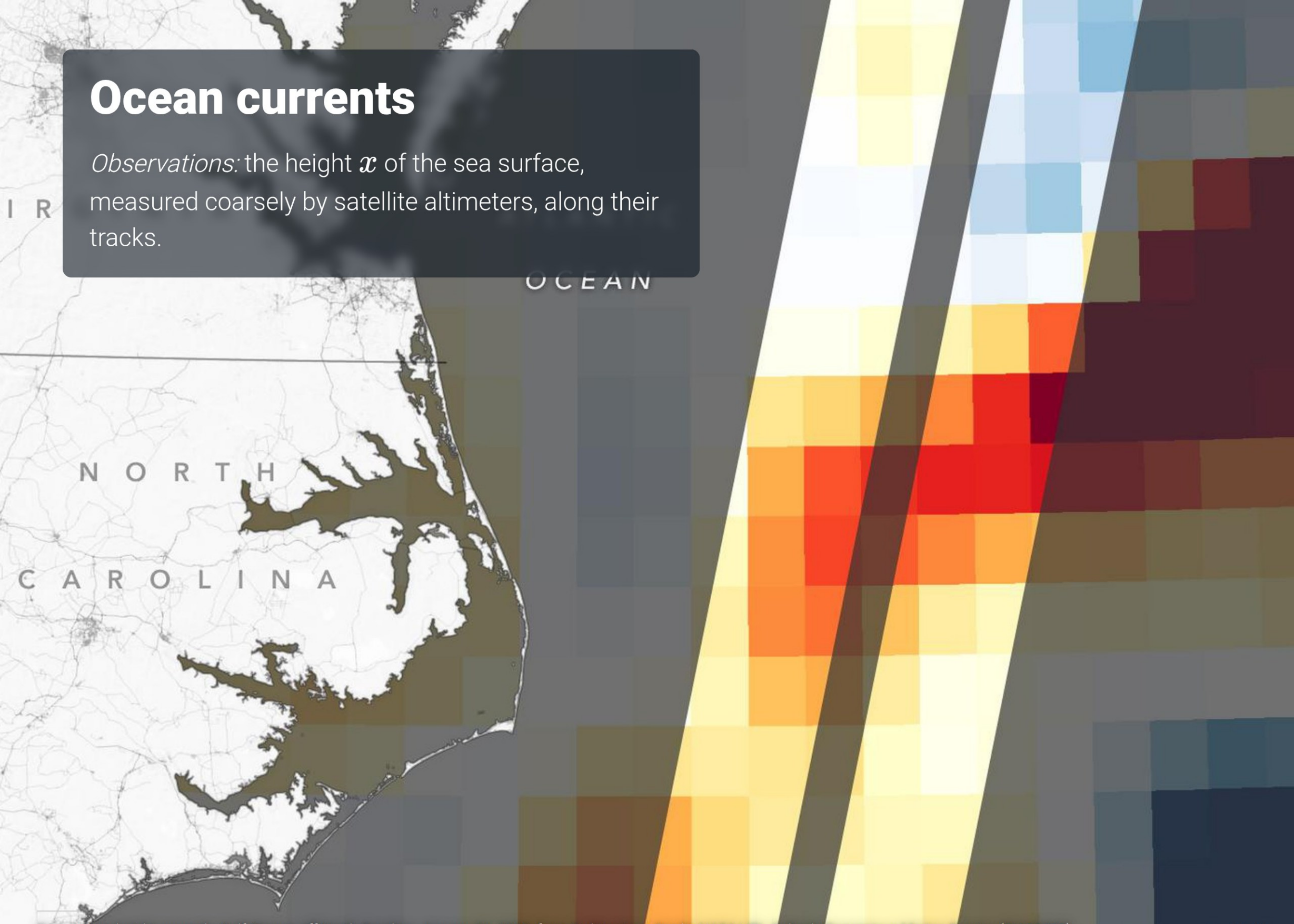
Max Dax
maximilian.dax@tuebingen.mpg.de

Exomints 2026
November 10, 2026

Illustration of WASP-39 b: NASA, ESA, CSA, J. Olmsted (STScI).

Ocean currents

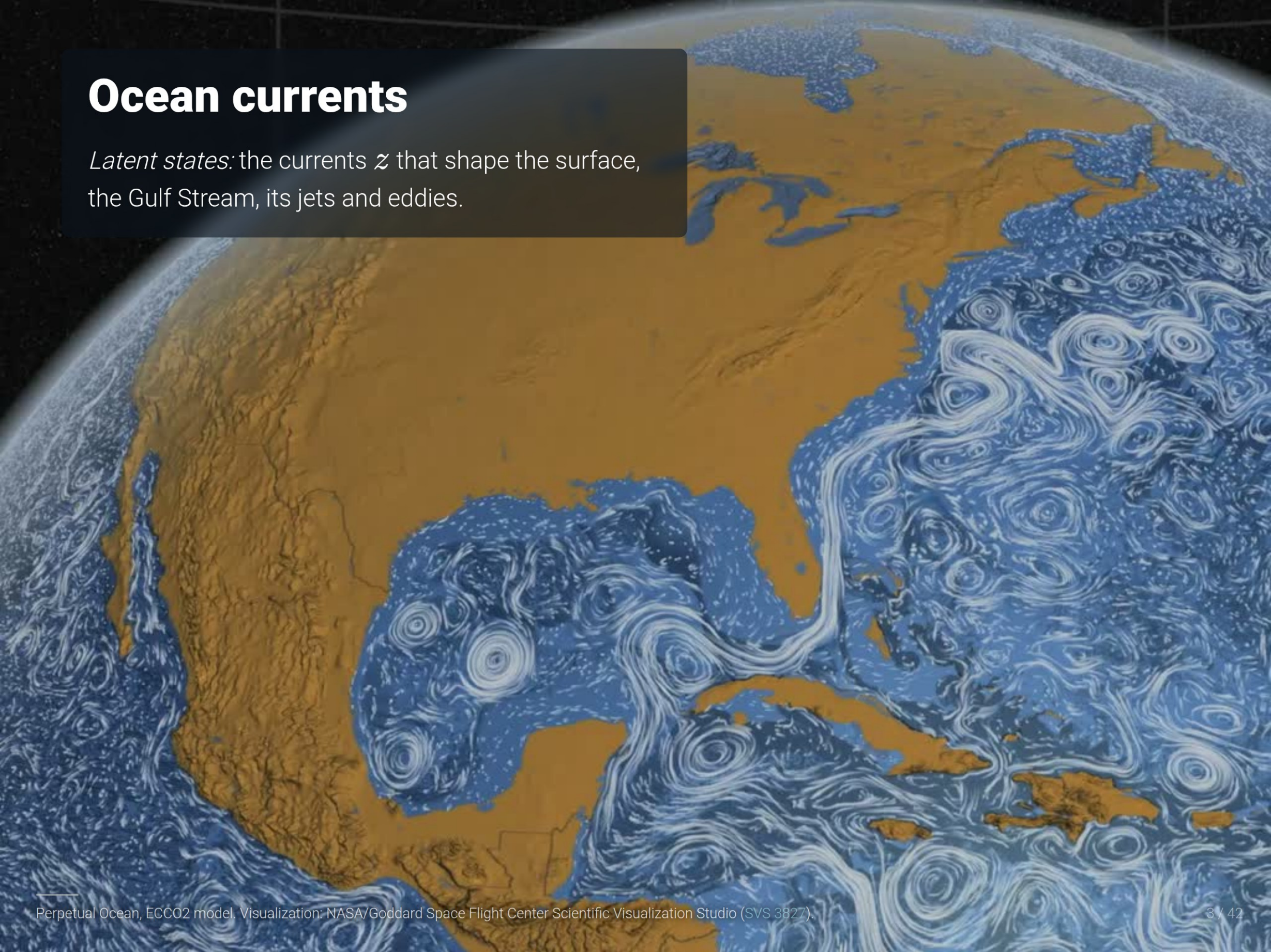
Observations: the height x of the sea surface, measured coarsely by satellite altimeters, along their tracks.



Sea surface height anomaly, Gulf Stream off North Carolina, January 21, 2023, from 7 altimeters. Credit: NASA/JPL-Caltech/Copernicus Marine Service (PIA25773).

Ocean currents

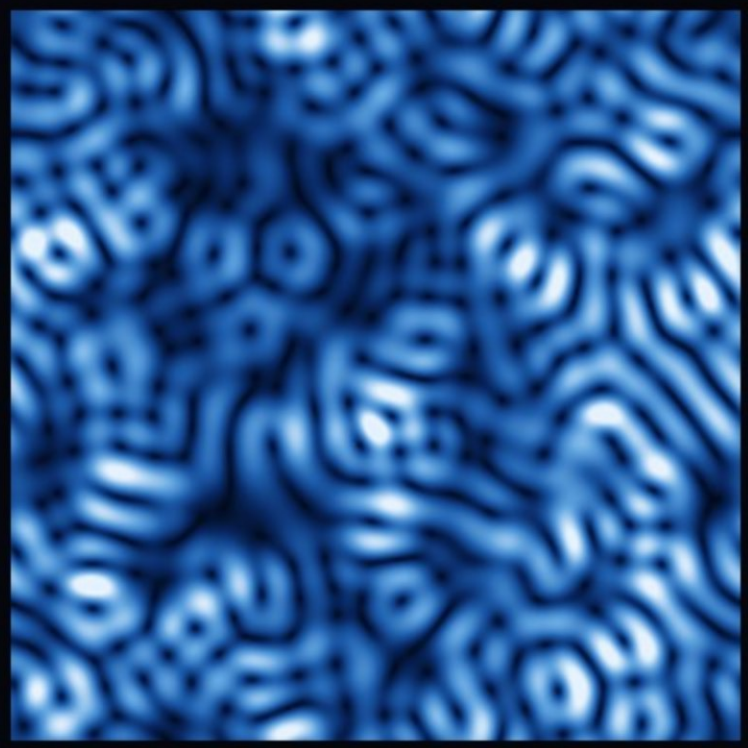
Latent states: the currents z that shape the surface, the Gulf Stream, its jets and eddies.



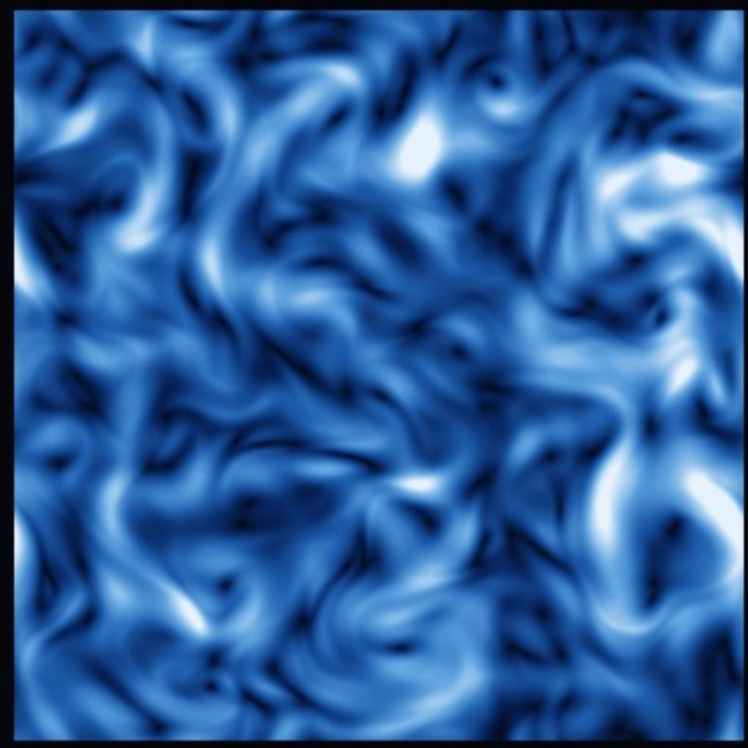
Perpetual Ocean, ECCO2 model. Visualization: NASA/Goddard Space Flight Center Scientific Visualization Studio (SVS 3827).

Ocean currents

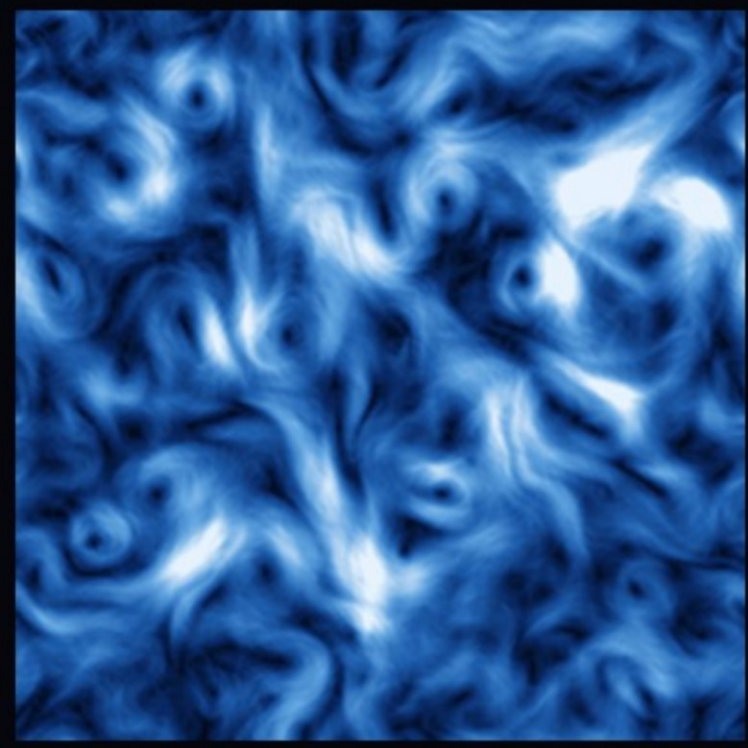
Parameters: the global constants θ of the ocean model, such as its viscosity ν.



$\nu = 10^{-2}$



$\nu = 10^{-3}$

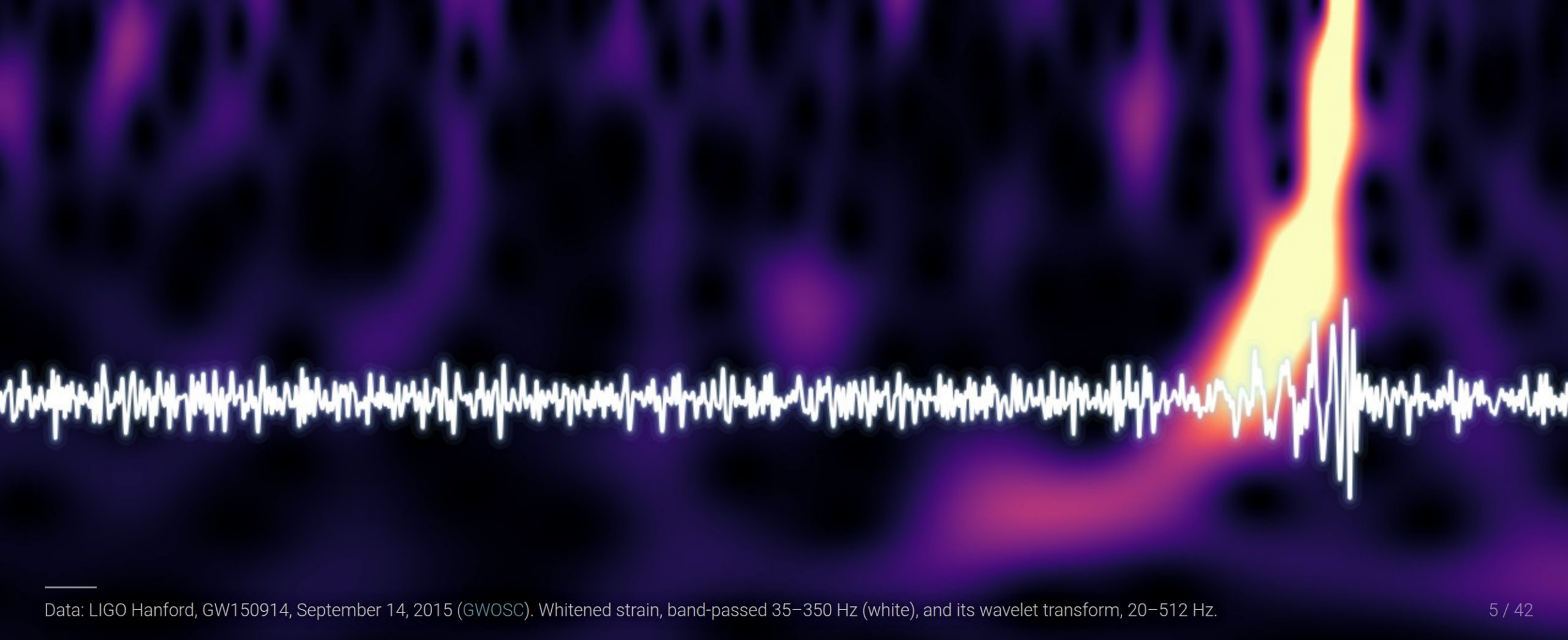


$\nu = 10^{-4}$

Illustration: 2D turbulence on a β-plane, same forcing, three viscosities; current speed.

Gravitational waves

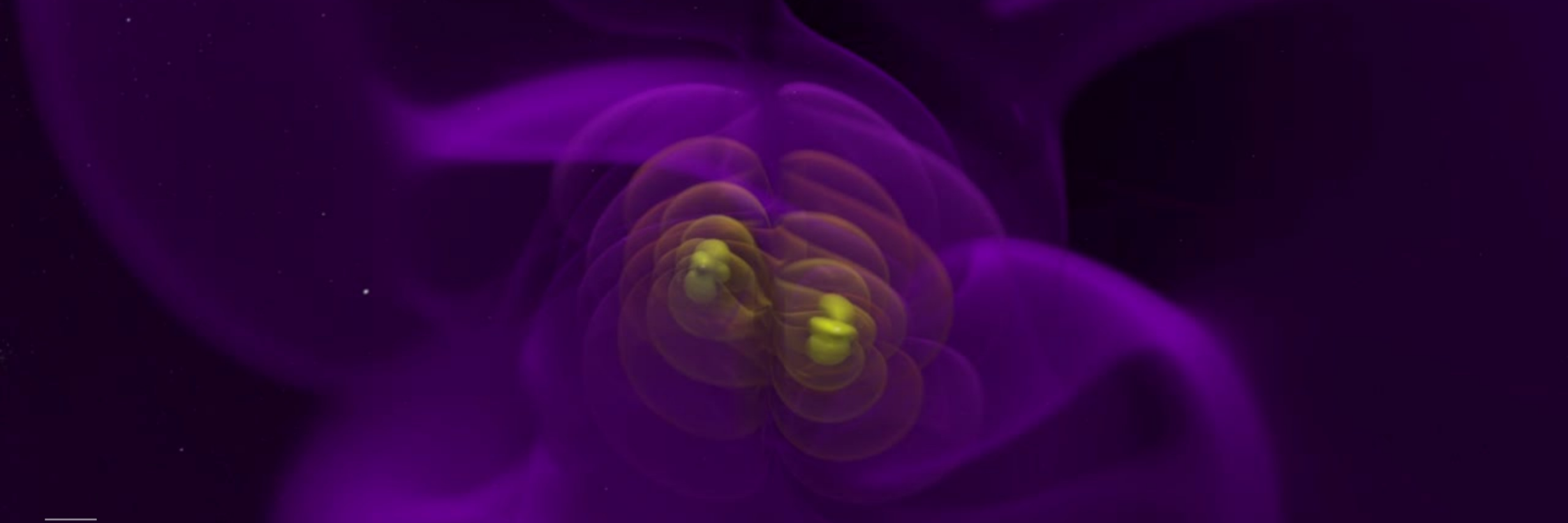
Observations: the strain x recorded by LIGO. In time and frequency, a chirp rises out of the noise.



Data: LIGO Hanford, GW150914, September 14, 2015 (GWOSC). Whitened strain, band-passed 35–350 Hz (white), and its wavelet transform, 20–512 Hz.

Gravitational waves

Latent states: the signal z of two black holes that spiral into each other, merge, and ripple space-time.



Simulation: B. J. Kelly, C. Henze, T. Sandstrom, NASA SVS 13197.

Gravitational waves

Parameters: the properties θ of the binary, masses, spins, distance, orientation, sky position (15 in total).

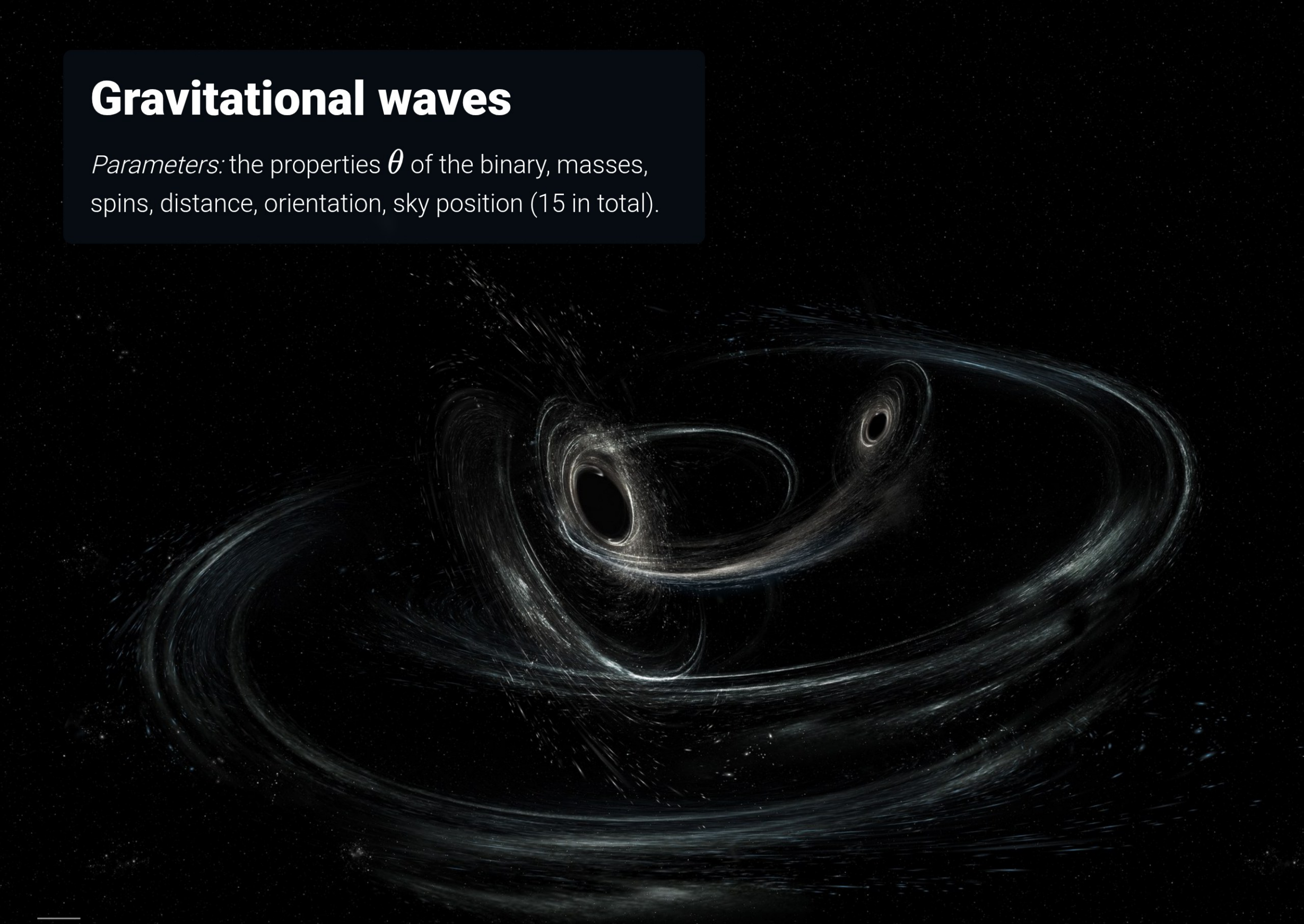
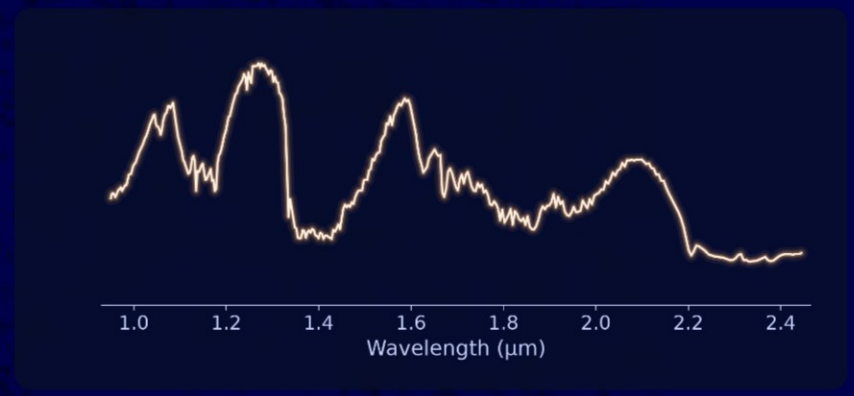


Illustration: LIGO/Caltech/MIT/Sonoma State (Aurore Simonnet).



Exoplanets

Observations: the spectrum $\boldsymbol{x}$ of a planet, whose light direct imaging separates from its star.

HR 8799, Keck, 2009–2016: Jason Wang (Caltech), Christian Marois (NRC Herzberg), CC BY 3.0. Spectrum: simulated benchmark of Vasist et al, A&A 2023.

Exoplanets

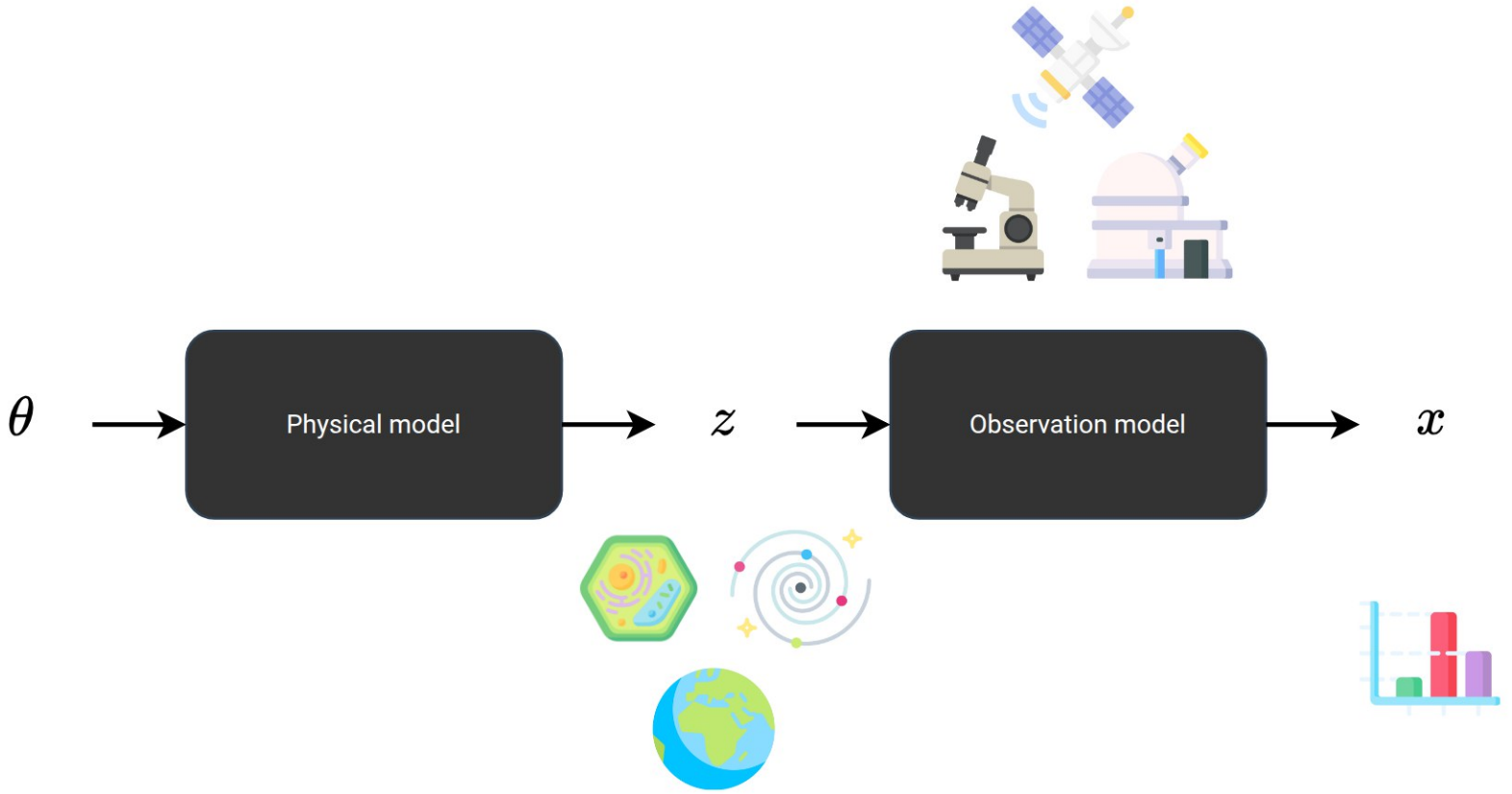
Latent states: the atmosphere z, its temperature profile, its molecules, its clouds.

VHS 1256 b. Illustration: NASA, ESA, CSA, Joseph Olmsted (STScI).

Exoplanets

Parameters: the bulk properties θ, C/O, metallicity, gravity, radius, clouds, mixing.

Ten hot Jupiters studied with Hubble (Sing et al, Nature 2016). Illustration: ESA/Hubble & NASA.



Inverse problems in science

Given an observation x, estimate the posterior distribution

$$p(\theta|x) = \frac{p(x|\theta)p(\theta)}{p(x)}, \quad \text{with} \quad p(x|\theta) = \int p(x|z)p(z|\theta)\mathrm{d}z,$$

of the parameters θ, with the latent states z integrated out.

Generation





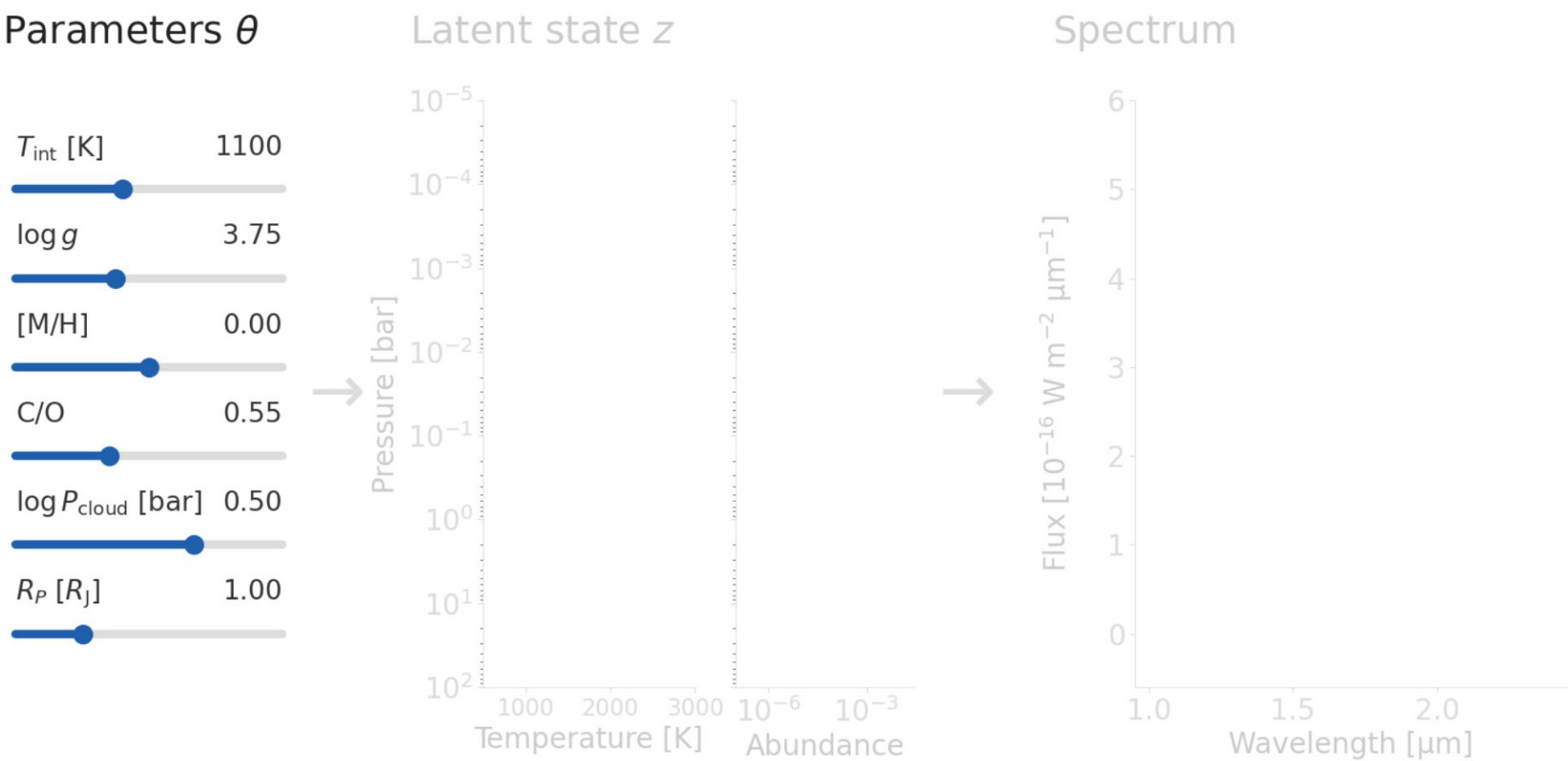


	Prior $p(\theta)$	**Physics** $p(z\|\theta)$	**Instrument** $p(x\|z)$
Ocean currents	viscosity, mixing, drag	Navier–Stokes	altimetry, noise
Gravitational waves	masses, spins, distance	general relativity	detector response, noise
Exoplanets	C/O, metallicity, clouds	radiative transfer	spectrograph, noise

Inference

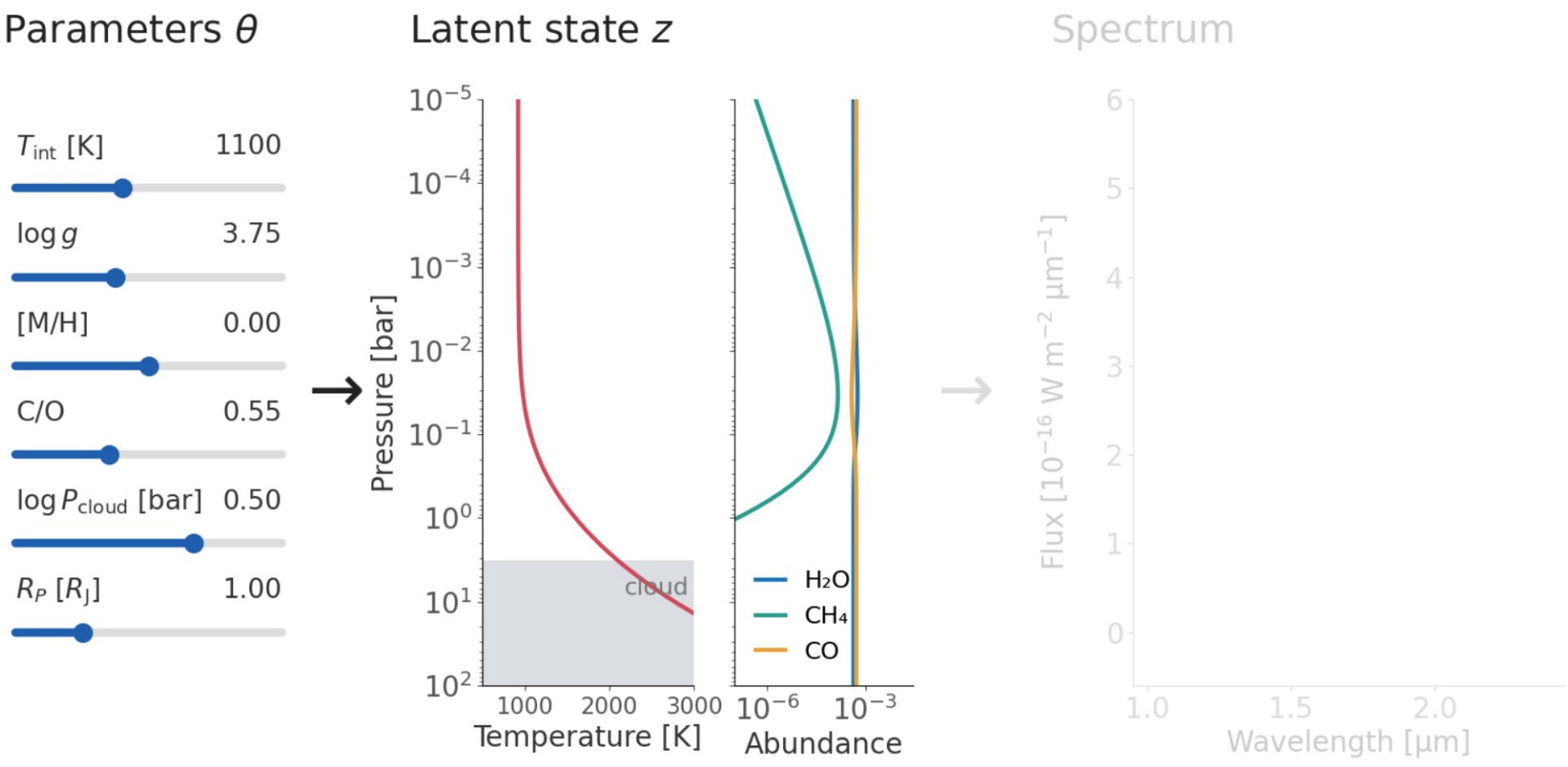
A minimal exoplanet simulator

Prior, $\theta \sim p(\theta)$: six parameters, each uniform over a range.



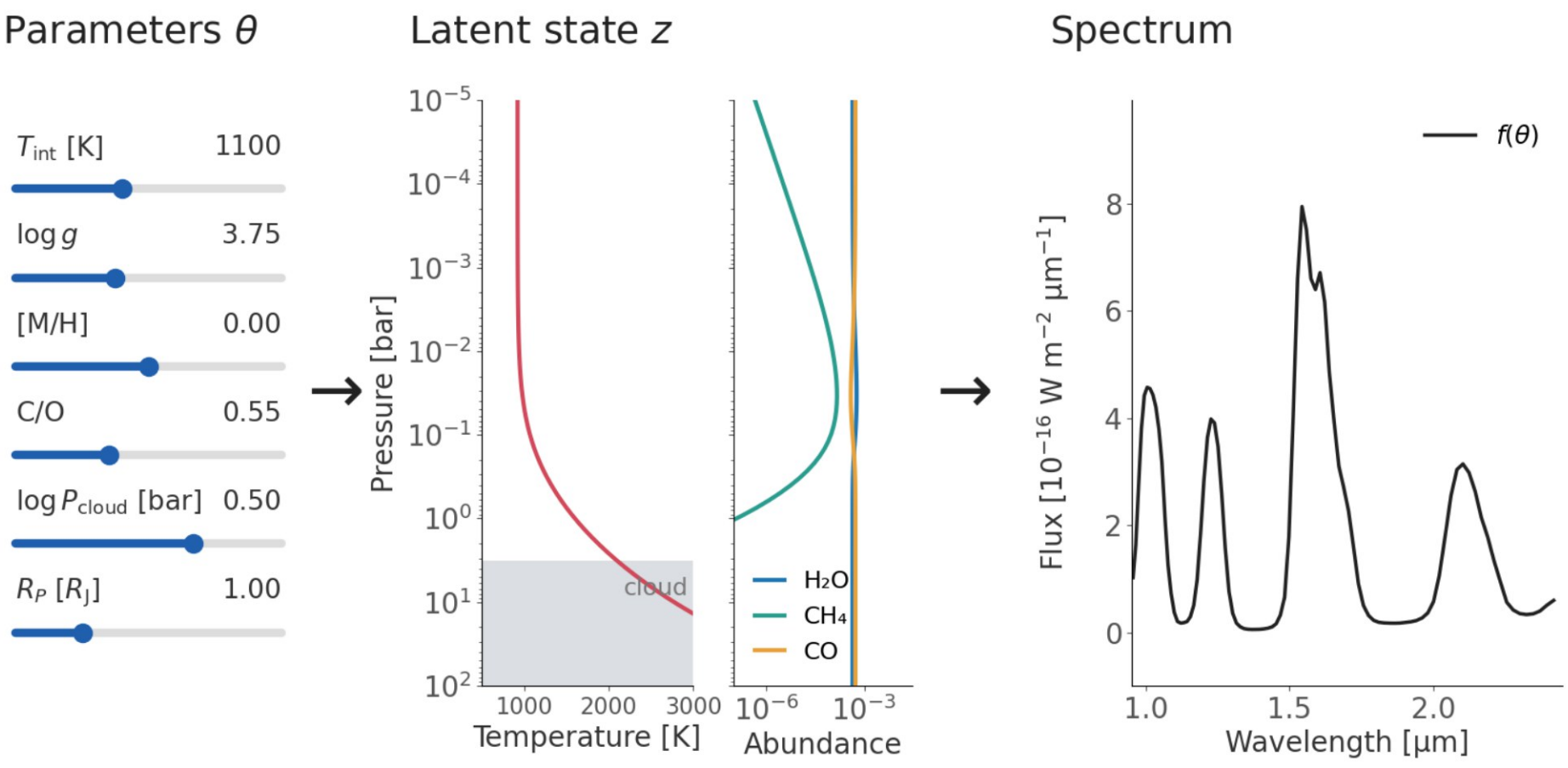
A minimal exoplanet simulator

Physics, $z \sim p(z|\theta)$: a temperature profile, chemistry, and a cloud deck.



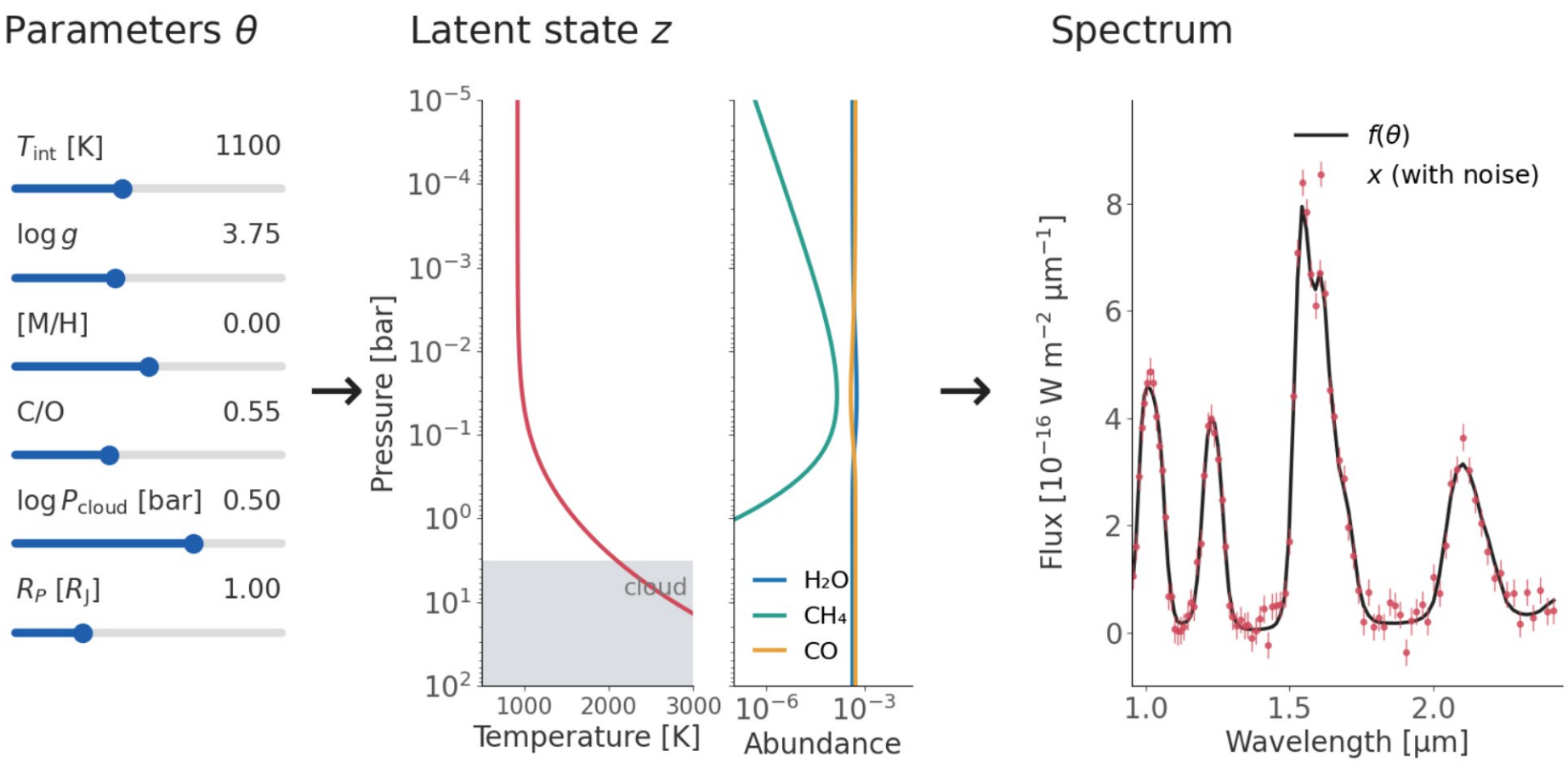
A minimal exoplanet simulator

Radiative transfer: the light $f(\theta)$ that the atmosphere emits.



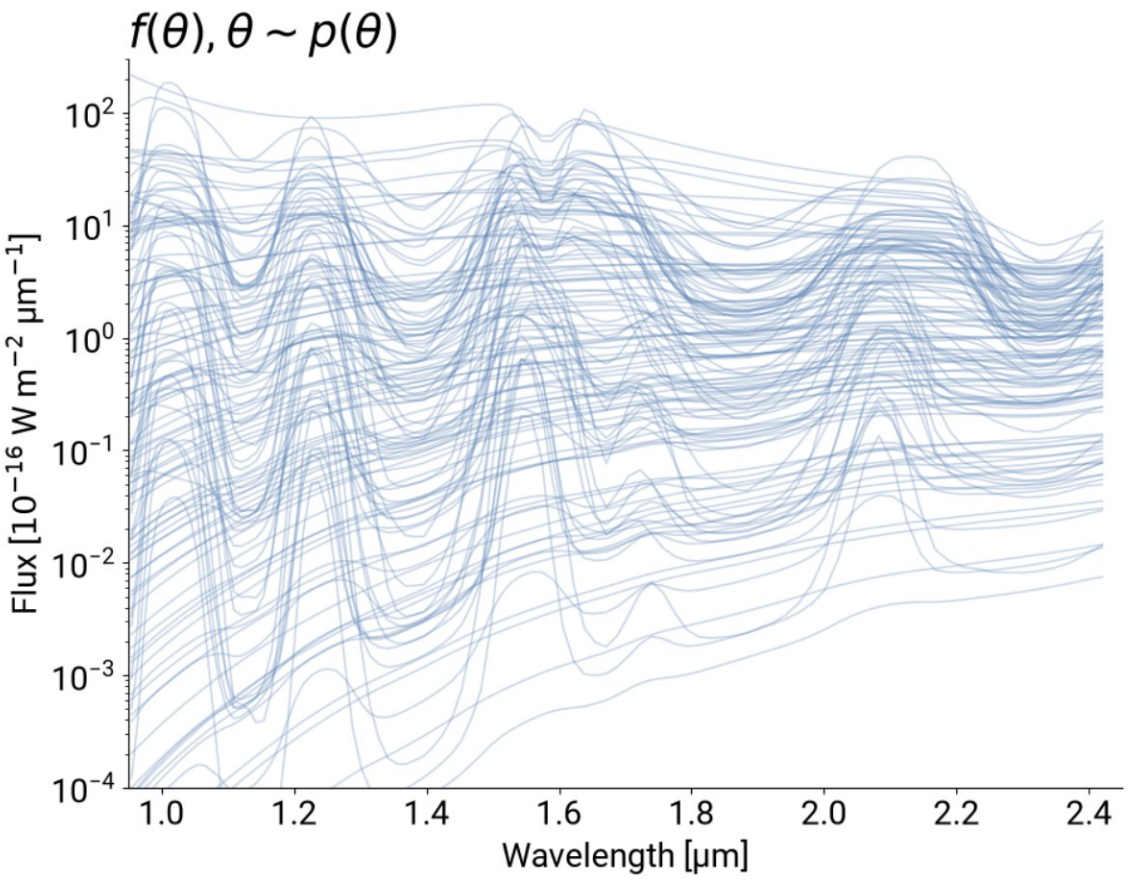
A minimal exoplanet simulator

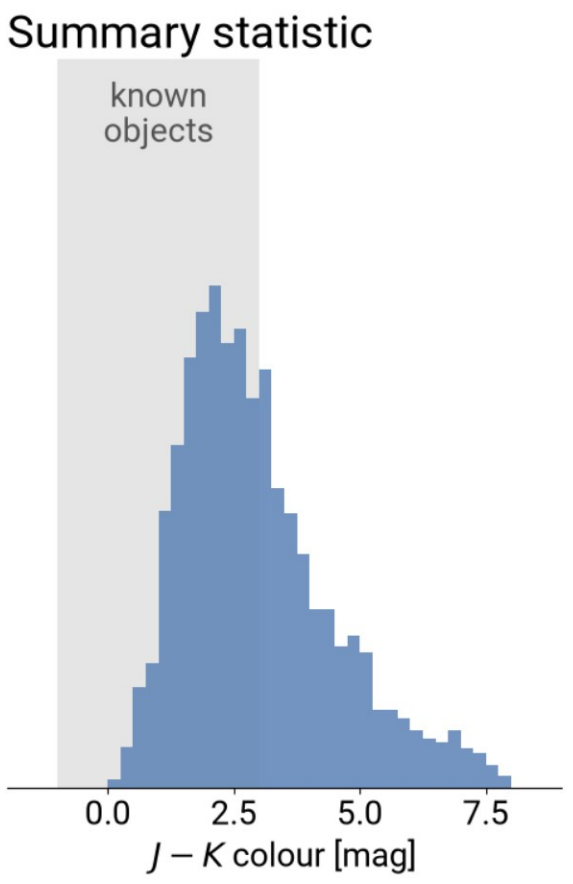
Instrument, $x \sim p(x|z)$: binned, with noise.

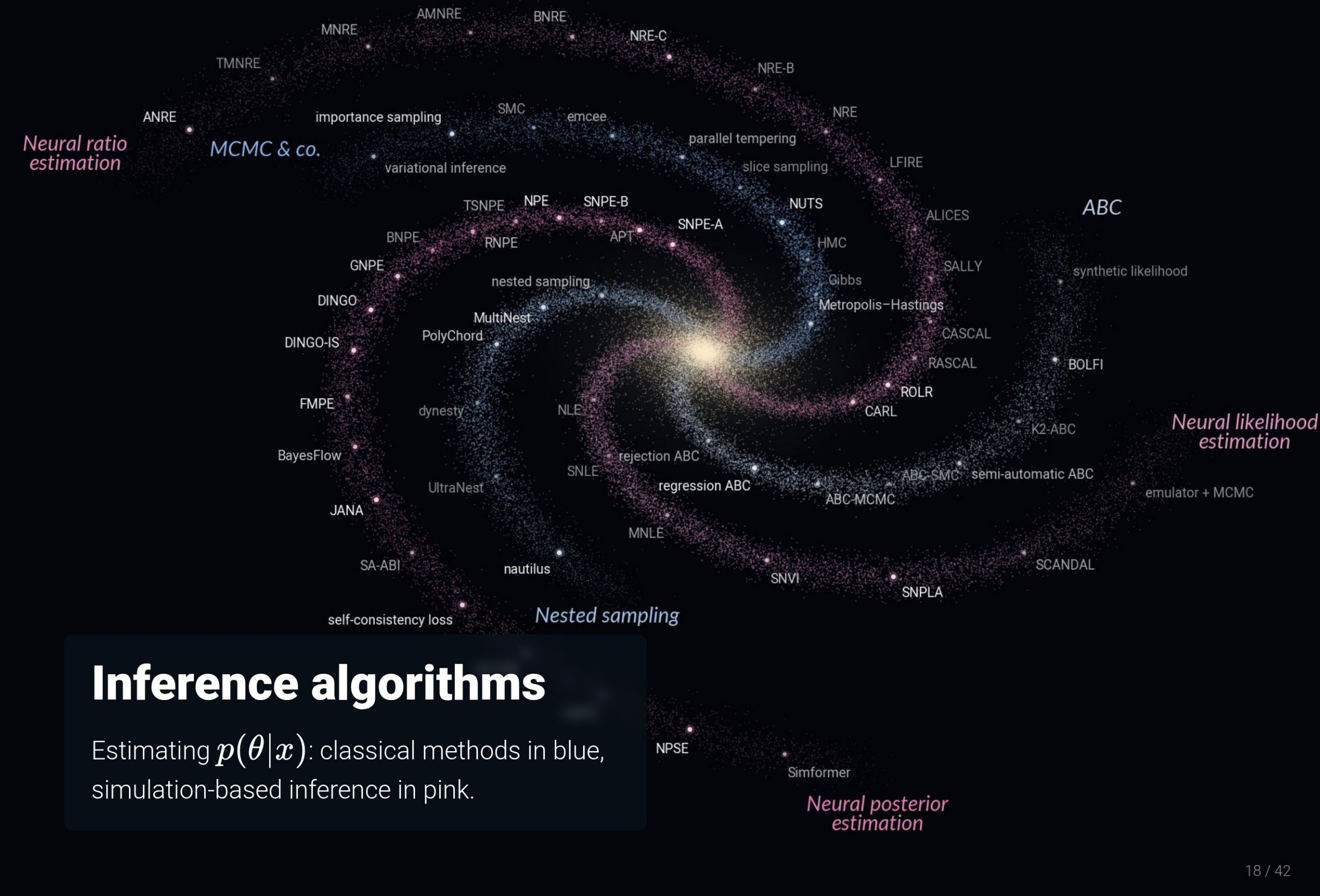


Prior predictive check

Before seeing any data, we check the **forward model** against domain knowledge: we draw $\theta \sim p(\theta)$, simulate, and compare summary statistics with known objects.

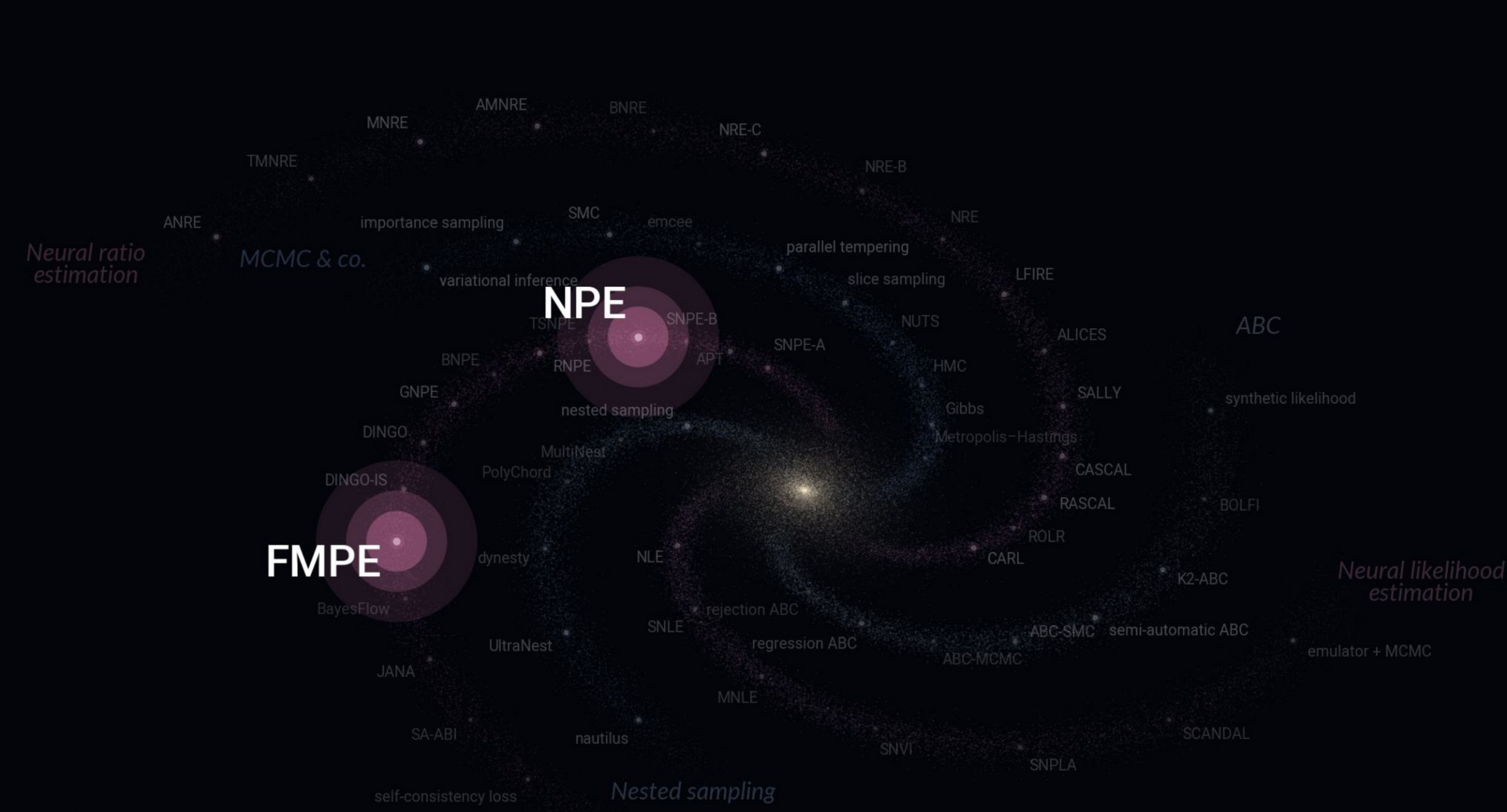






Inference algorithms

Estimating $p(\theta|x)$: classical methods in blue, simulation-based inference in pink.



Inference algorithms

Estimating $p(\theta|x)$: classical methods in blue, simulation-based inference in pink.

Neural posterior estimation

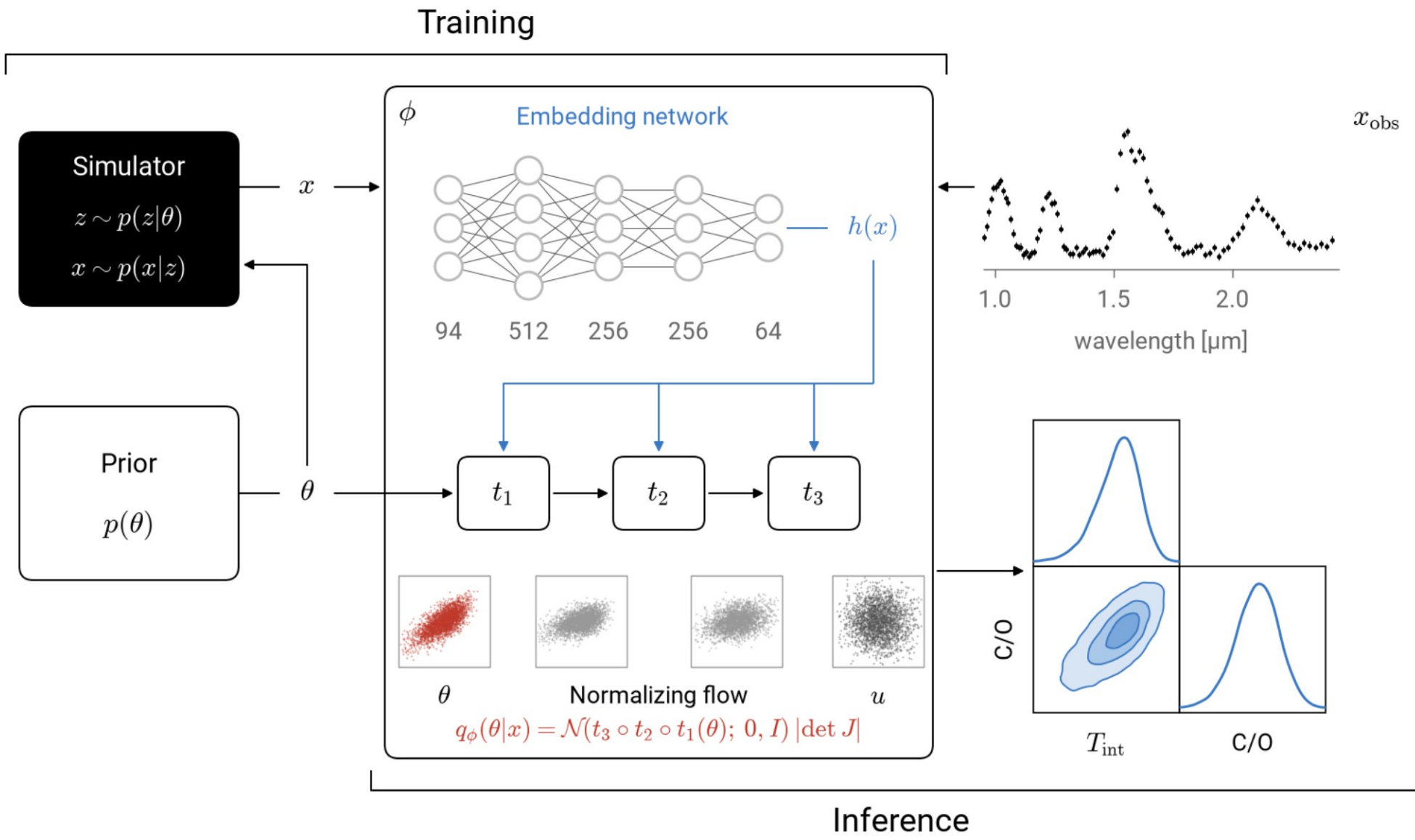
We train a conditional density estimator $q_\phi(\theta|x)$, typically a **normalizing flow**, to approximate the posterior of **any** observation x,

$$\phi^\star = \arg\min_\phi \; \mathbb{E}_{p(\theta,x)} \left[-\log q_\phi(\theta|x) \right].$$

The expectation runs over pairs (θ, x) from the simulator, so that the likelihood is never evaluated.

Up to a constant, this loss is the expected divergence to the true posterior,

$$\mathbb{E}_{p(x)} \, \mathbf{KL}\big(p(\theta|x) \,\|\, q_\phi(\theta|x)\big).$$



By the change of variables, the density is exact,

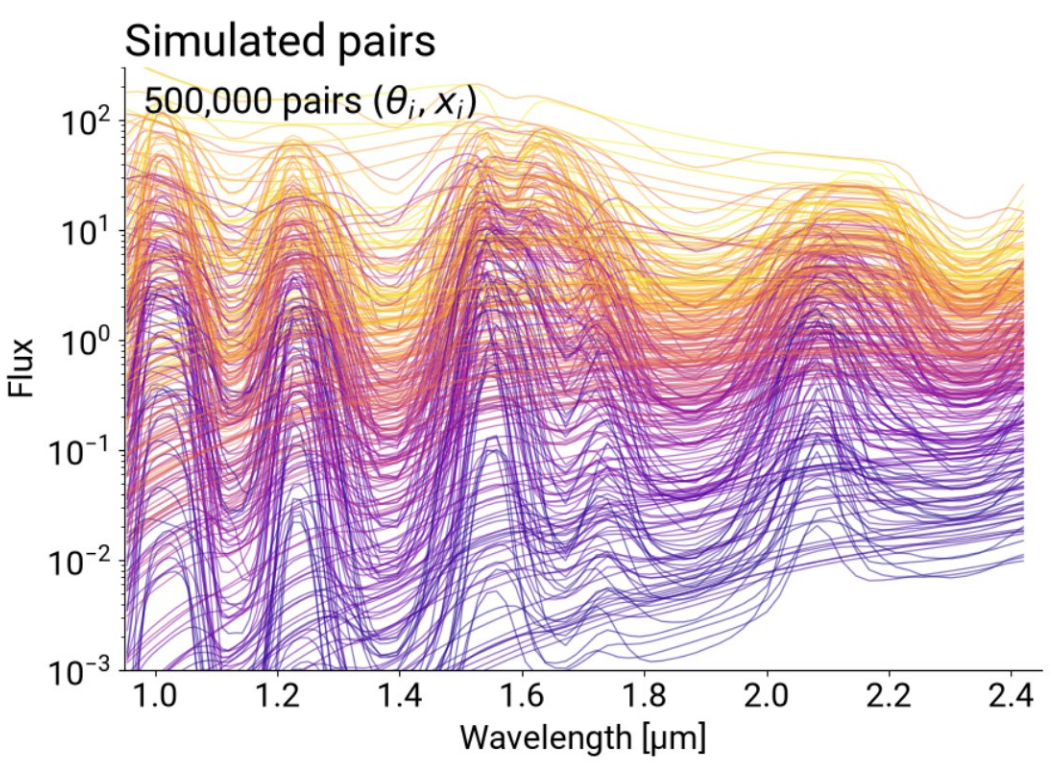
$$\log q_\phi(\theta|x) = \log \mathcal{N}(t_3 \circ t_2 \circ t_1(\theta); 0, I) + \sum_{k=1}^{3} \log \left|\det J_{t_k}\right| .$$

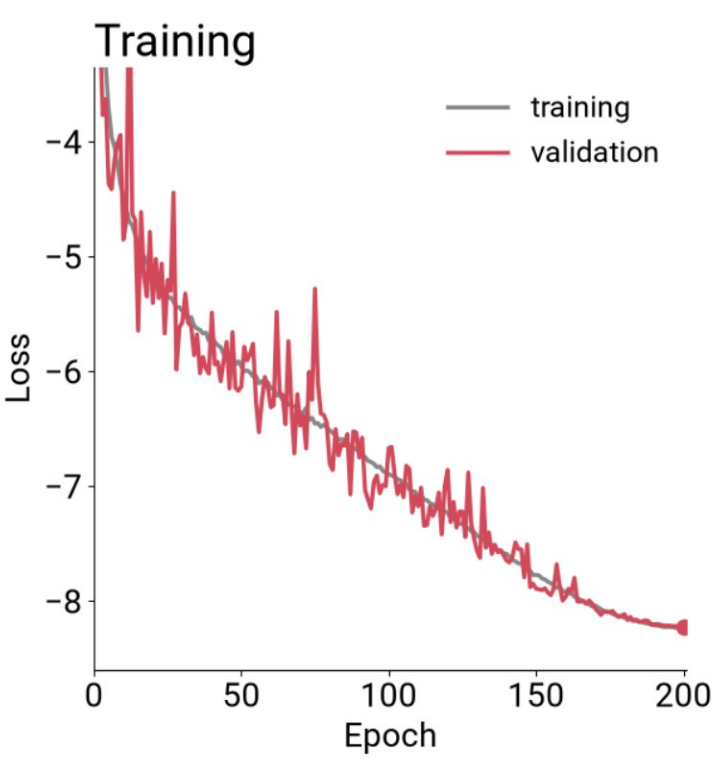
Training

We simulate N pairs $(\theta_i, x_i) \sim p(\theta, x)$ and minimize the empirical estimate of the loss,

$$L(\phi) = \mathbb{E}_{p(\theta,x)} \left[-\log q_\phi(\theta|x) \right] \approx \frac{1}{N} \sum_{i=1}^{N} -\log q_\phi(\theta_i|x_i),$$

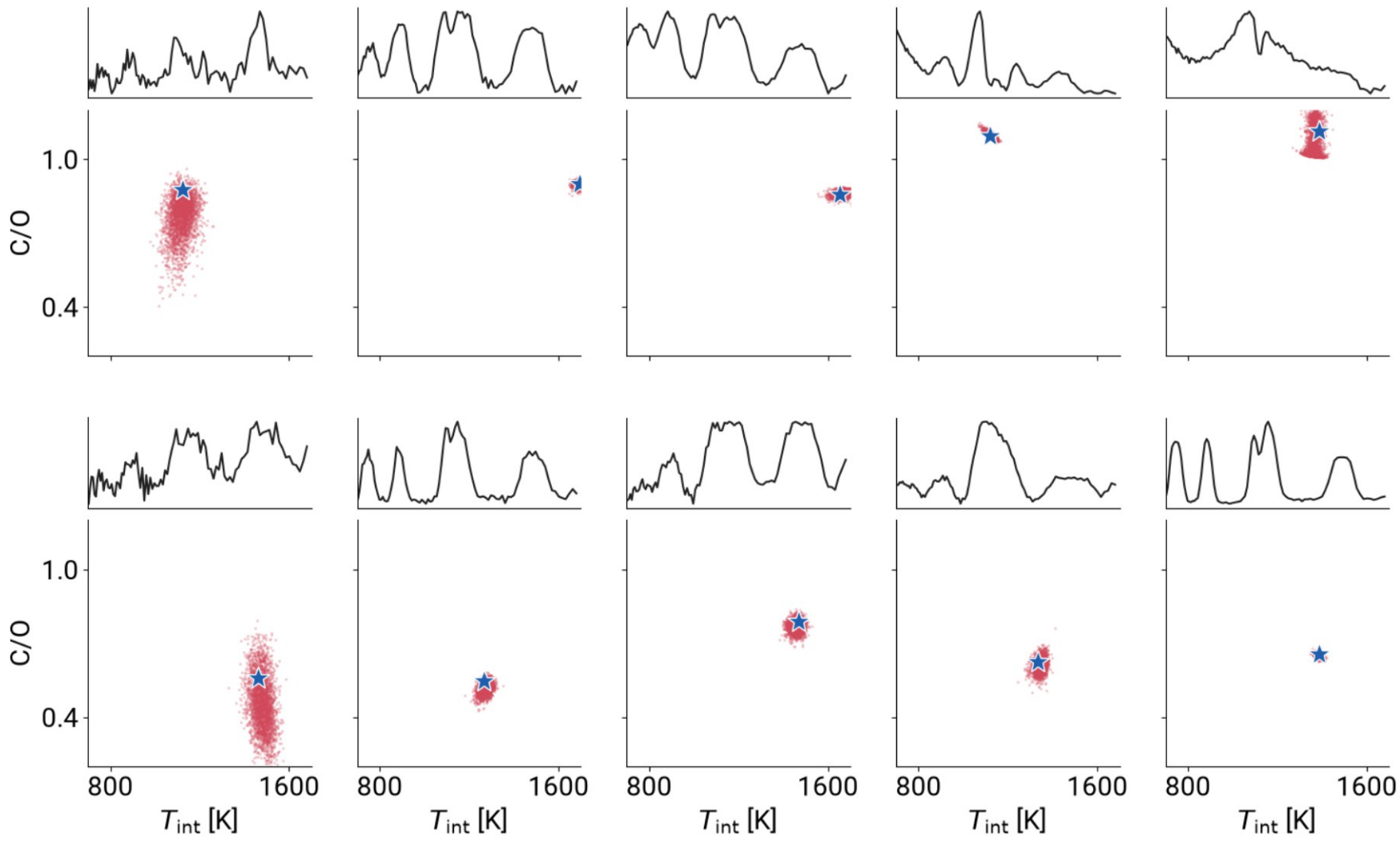
with the noise added on the fly.





Amortization

Once trained, the network gives the posterior of any new spectrum in a fraction of a second.

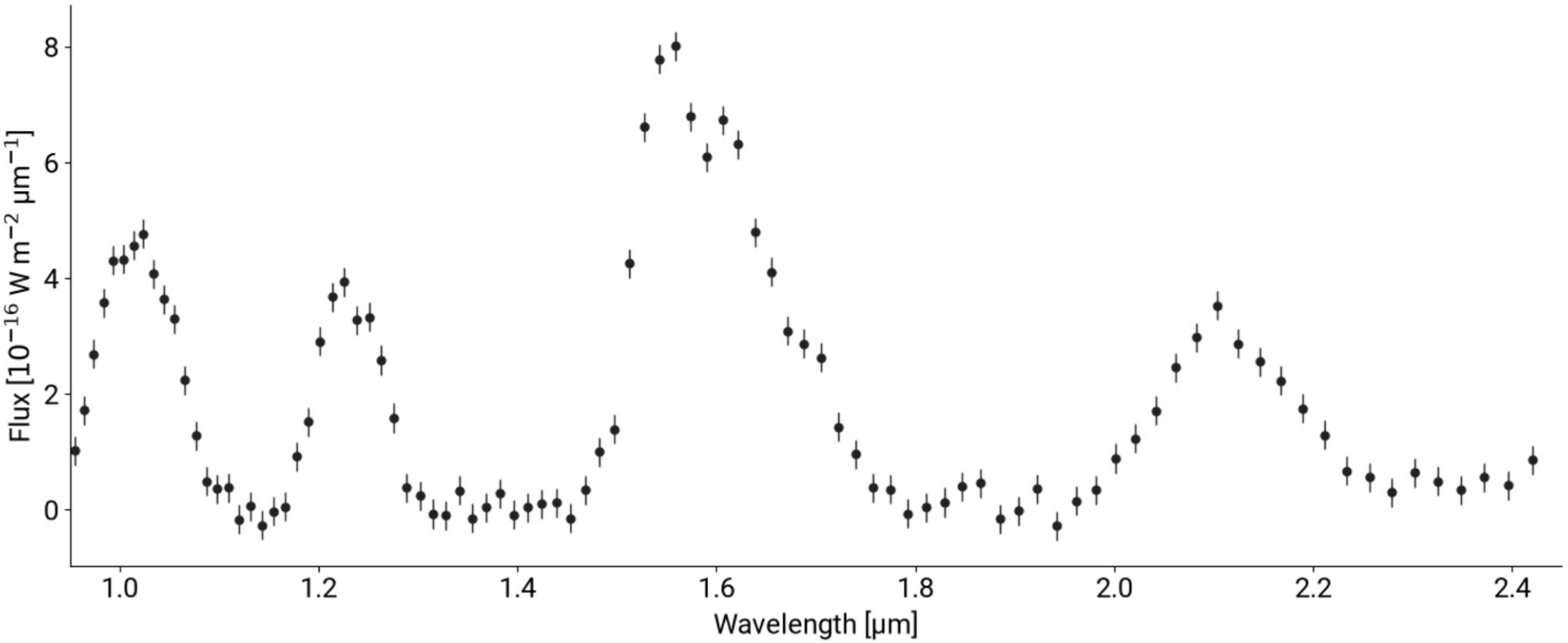


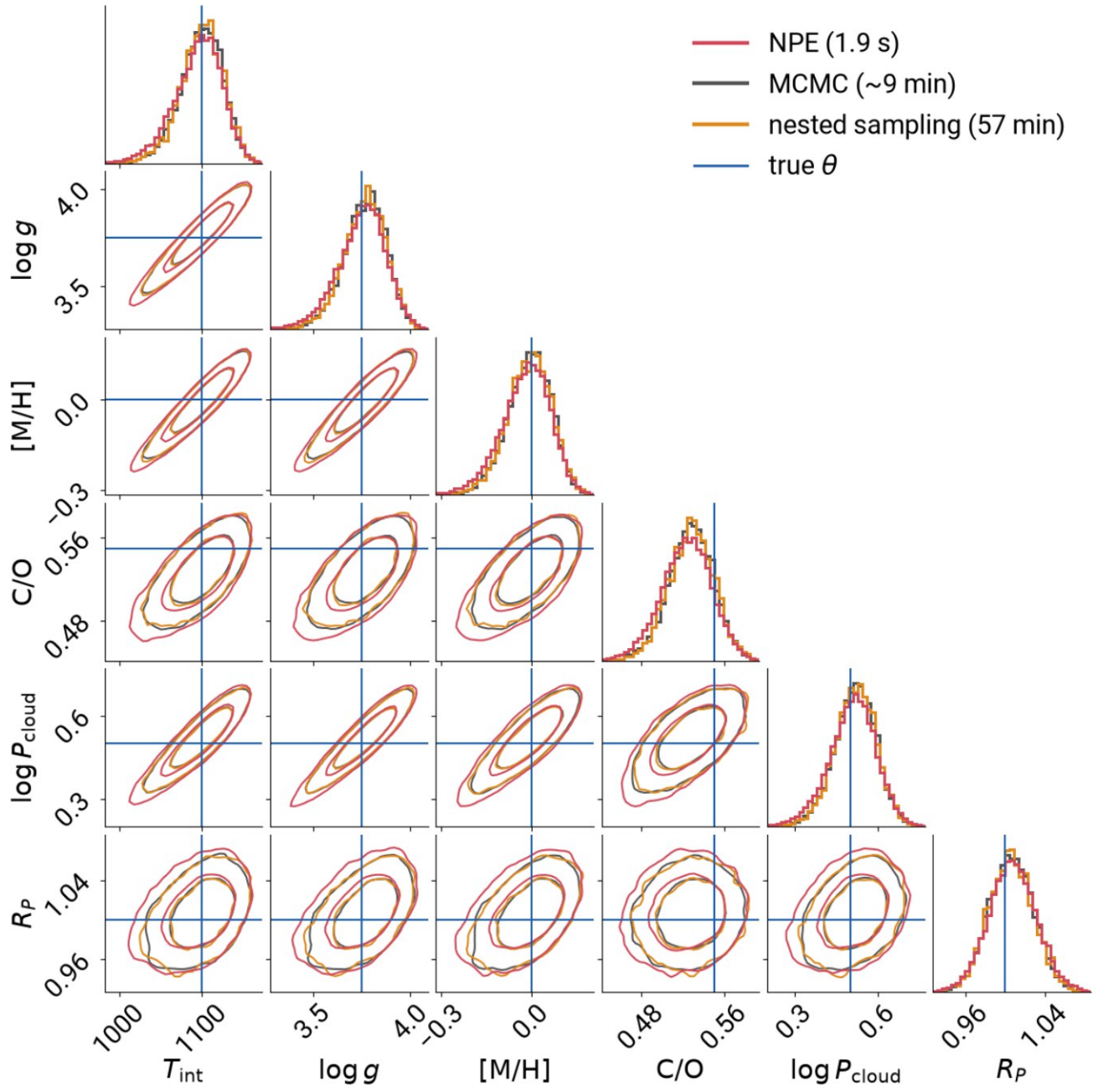
NPE: 1.9 s for 10 spectra

MCMC or nested sampling: ~90 min

Results on the toy problem

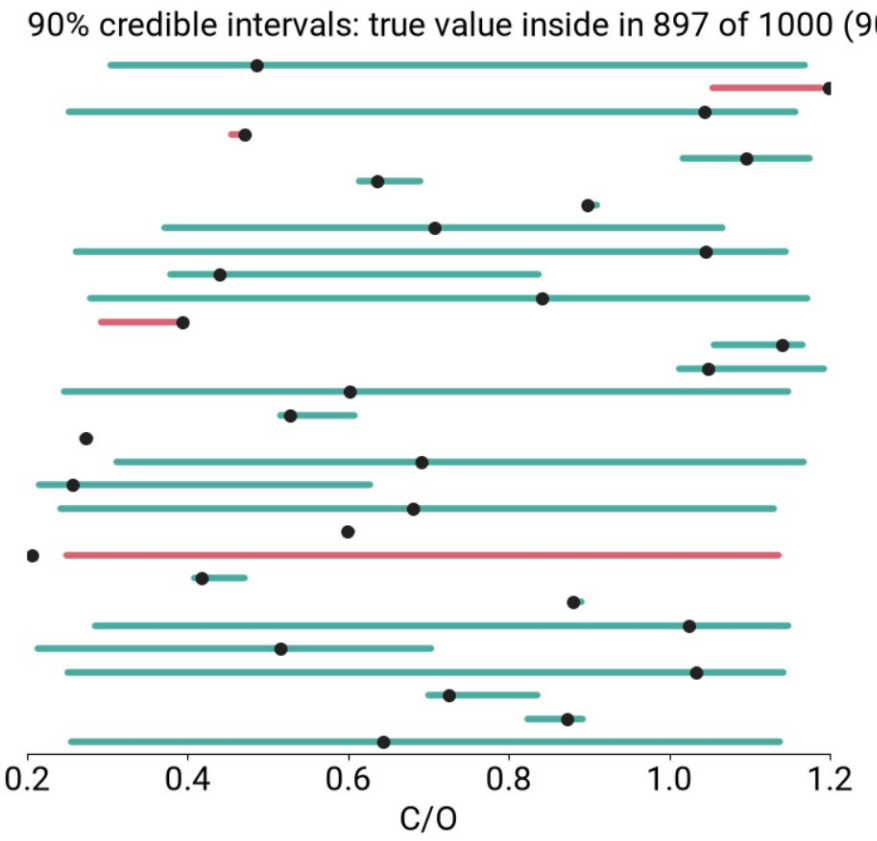
Consider an observed spectrum x_{obs}, simulated from known parameters $\theta^\star$ hidden from the methods.

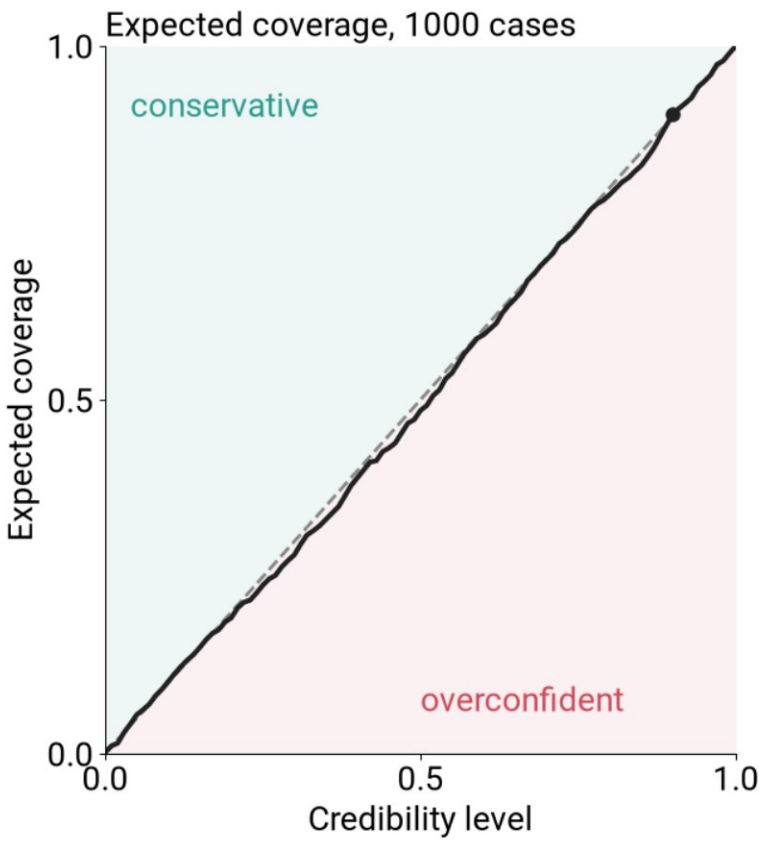



NPE (1.9 s)
MCMC (~9 min)
nested sampling (57 min)
true θ
log g
[M/H]
C/O
log P_cloud
R_P
T_int
4.0
3.5
0.0
−0.3
0.56
0.48
0.6
0.3
1.04
0.96
1000
1100

Calibration

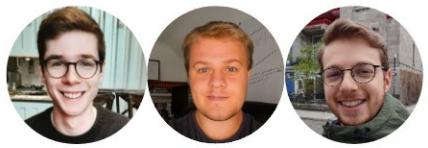
A posterior is **calibrated** if its 90% credible regions contain the true parameters 90% of the time, and likewise at every level.



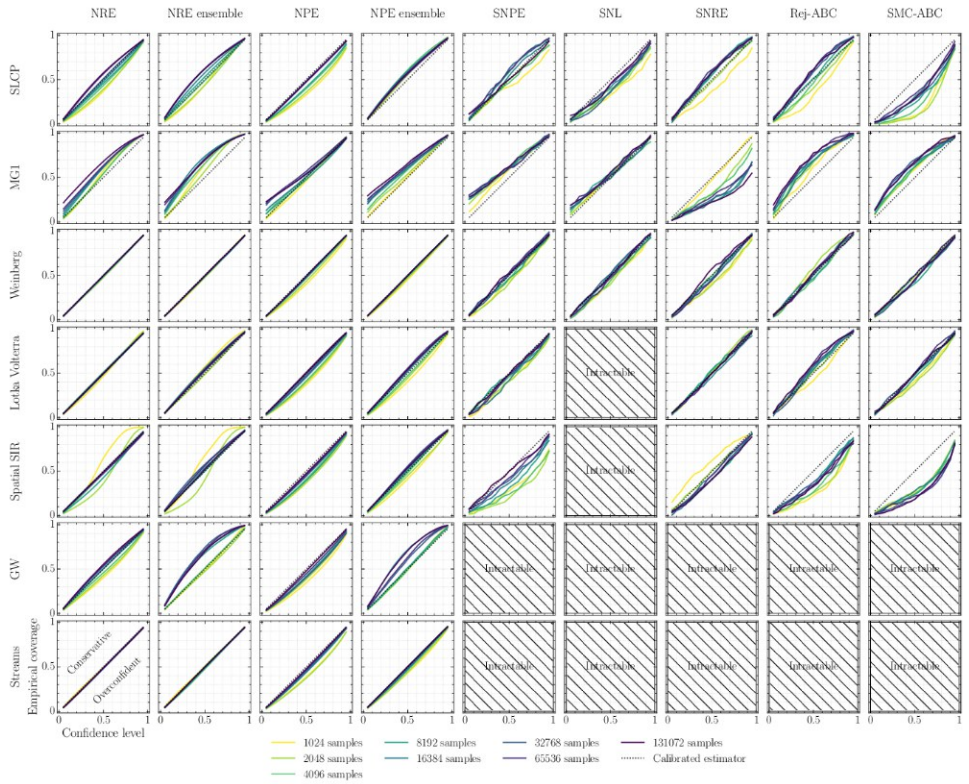


Coverage is necessary but not sufficient, since the prior itself is calibrated. TARP, MIRA and L-C2ST are stronger tests.

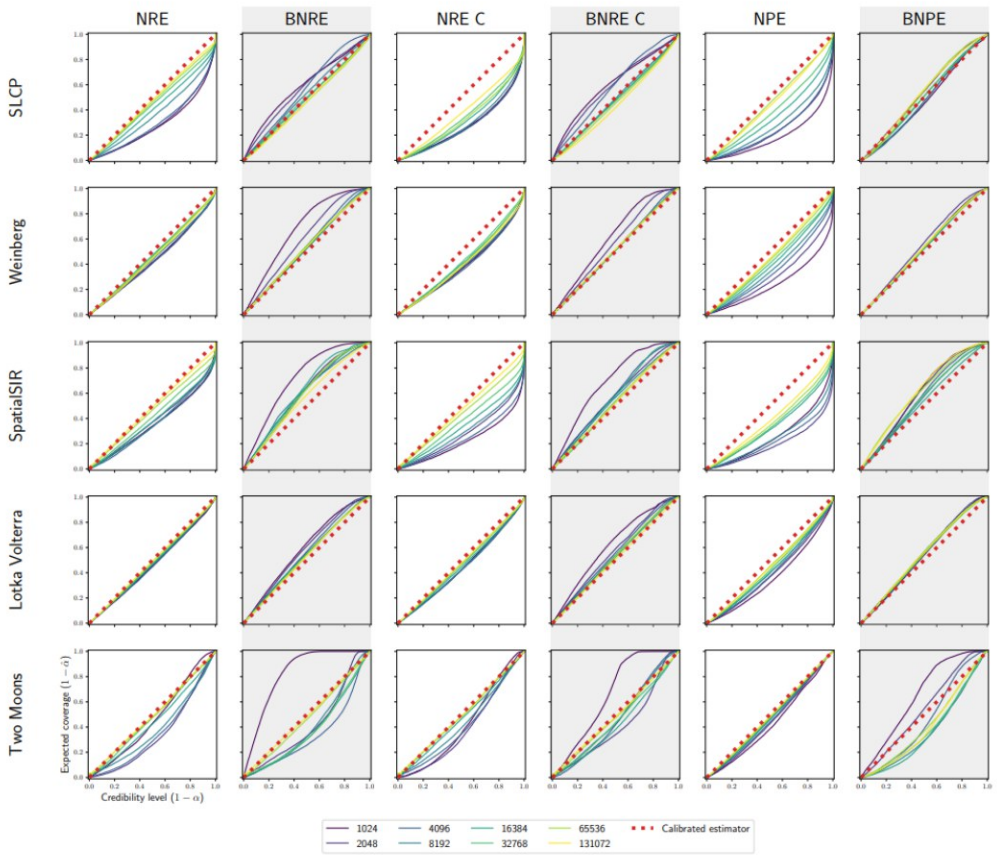
TARP: Lemos et al, ICML 2023. MIRA: Sharief et al, ICML 2026. L-C2ST: Linhart et al, NeurIPS 2023.



Neural posteriors can be **overconfident**, even when trained as recommended.
Balancing the training makes them conservative, which is preferable for science.



Coverage of common SBI algorithms

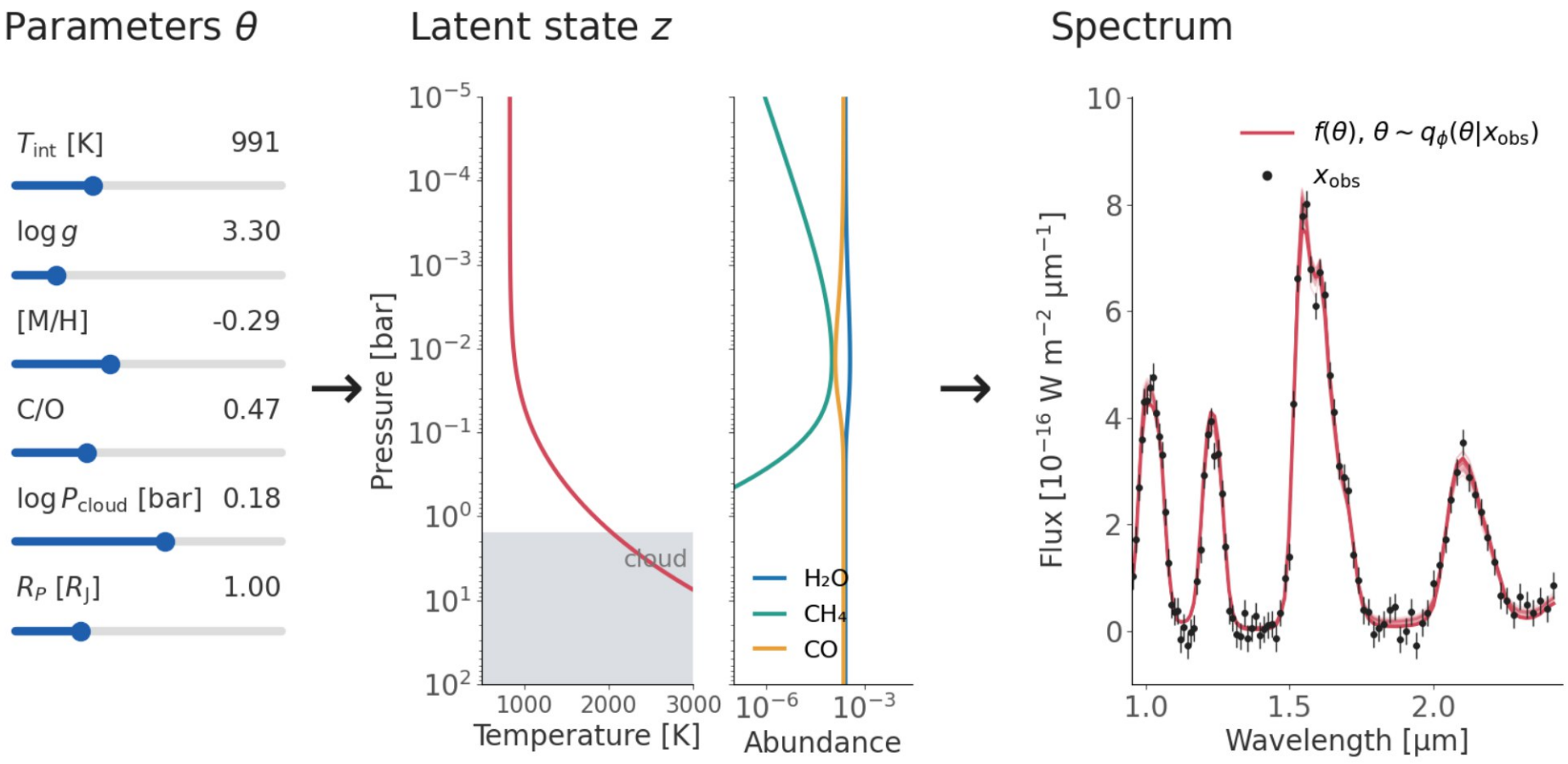


Balanced NPE (last column)

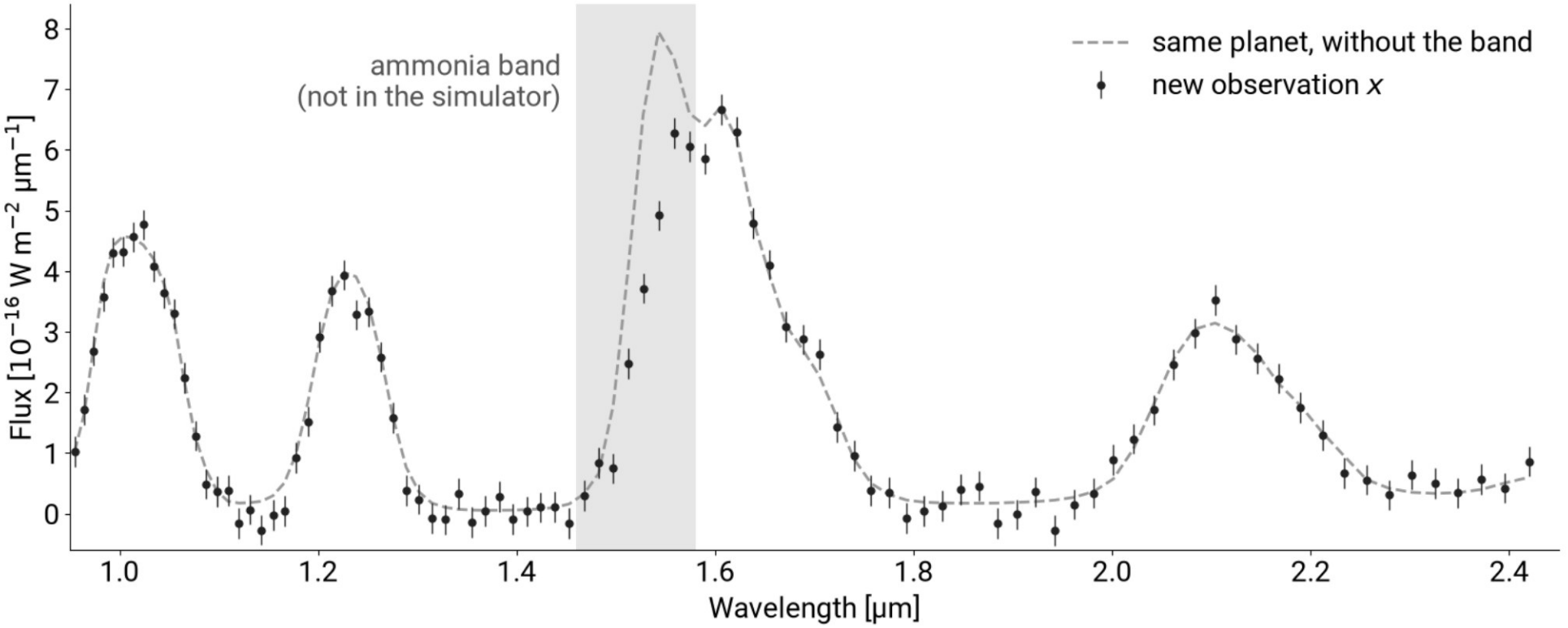
Hermans et al, TMLR 2022. Delaunoy et al, 2023. See also Falkiewicz et al, NeurIPS 2023.

Posterior predictive check

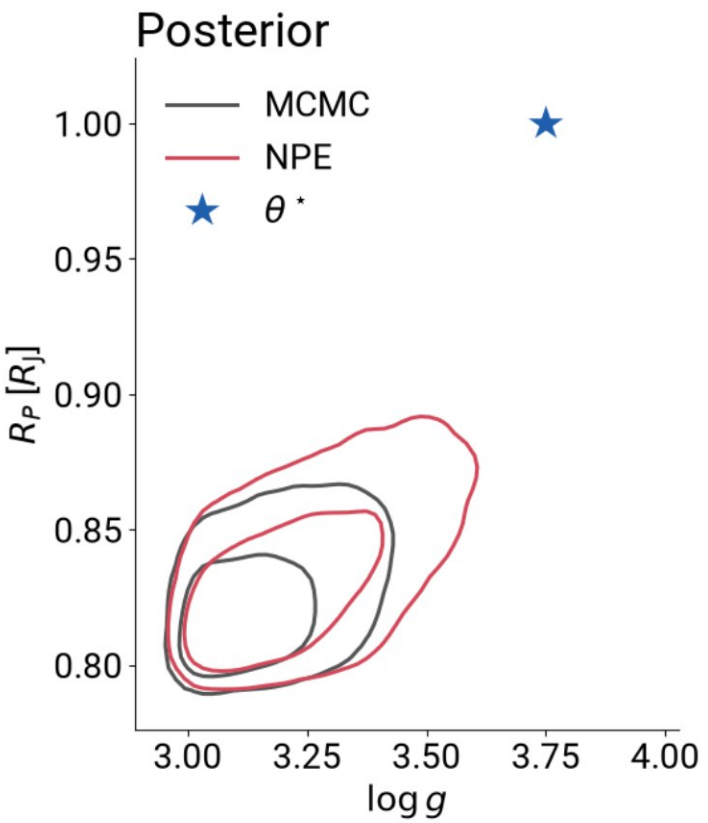
To check the **forward model**, we draw $\theta \sim q_\phi(\theta|x_{\text{obs}})$, simulate, and compare with the observation.

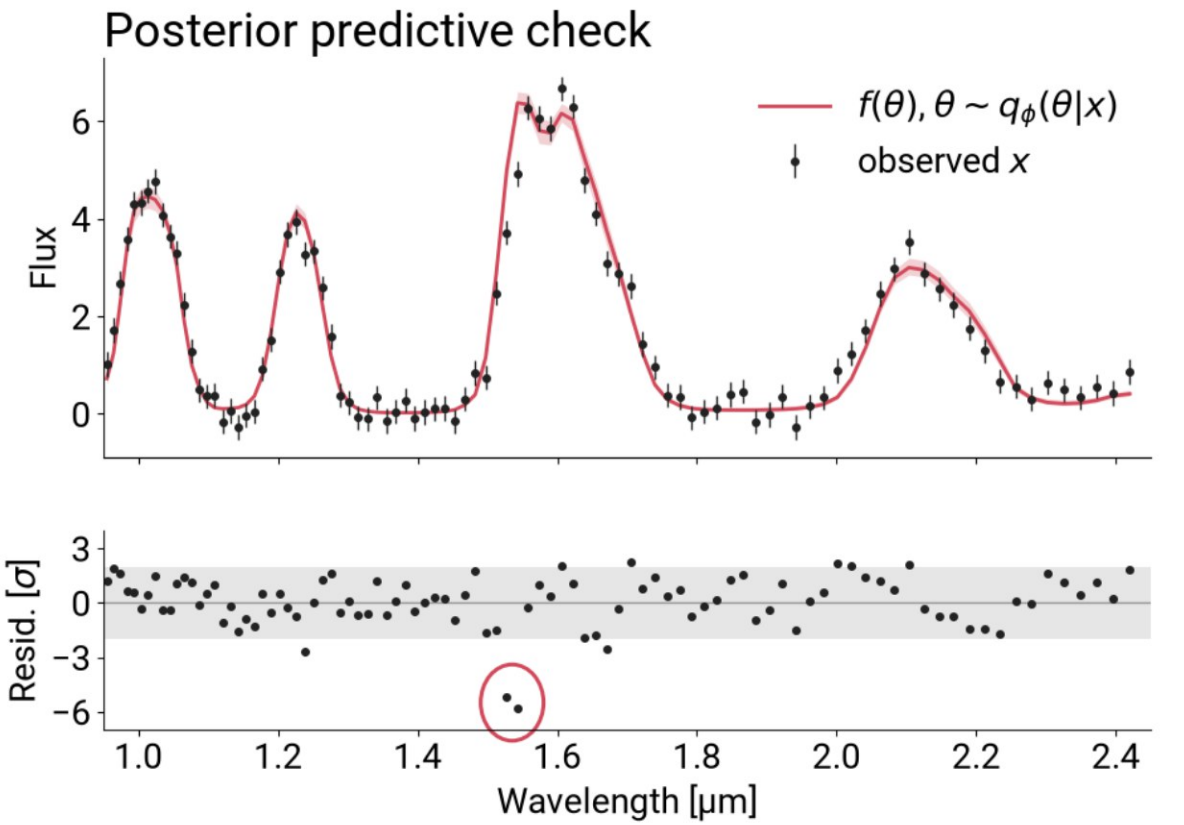


When the model is wrong



Consider now a new observation of the same planet, with an absorption band at 1.52 µm that the simulator does not have.





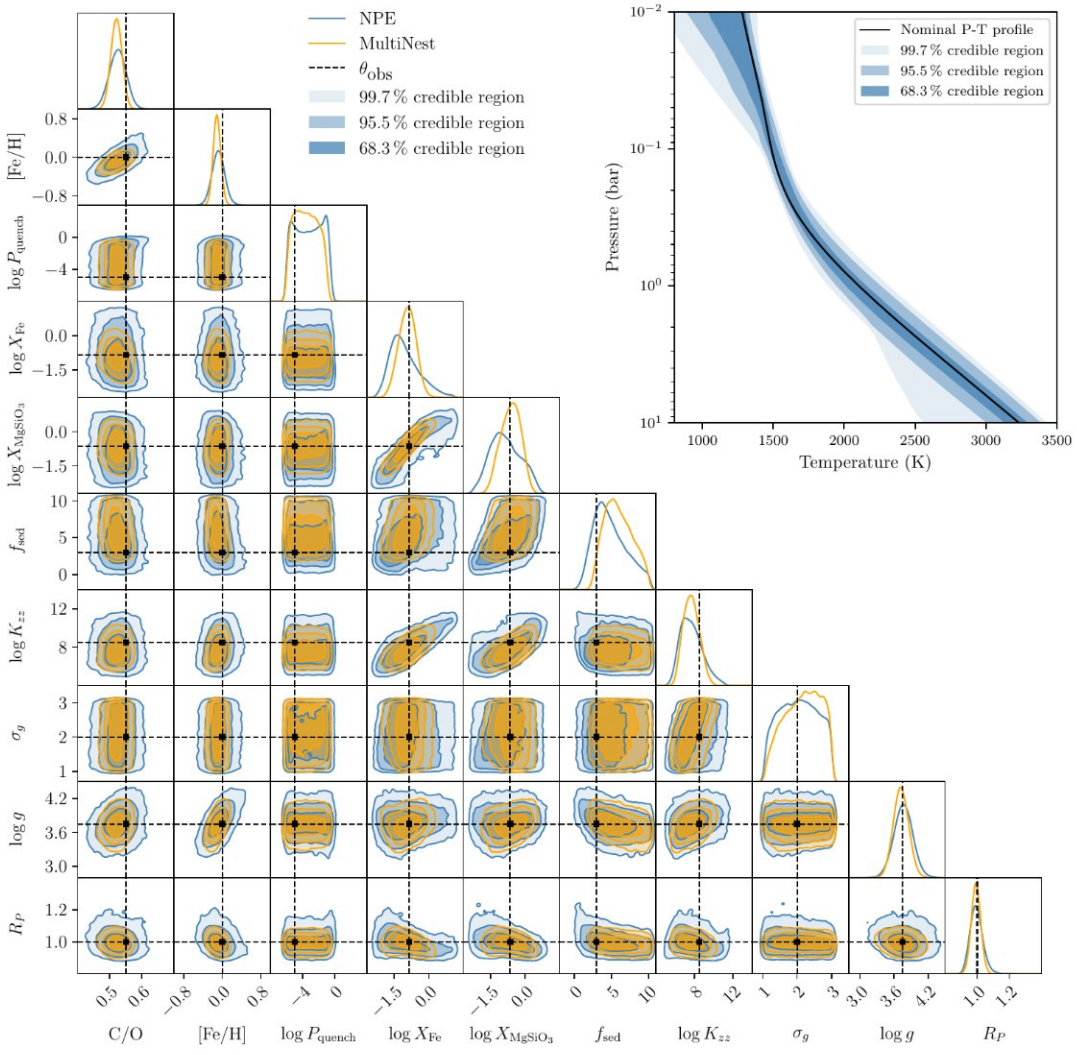
Both posteriors are **confident and wrong**.

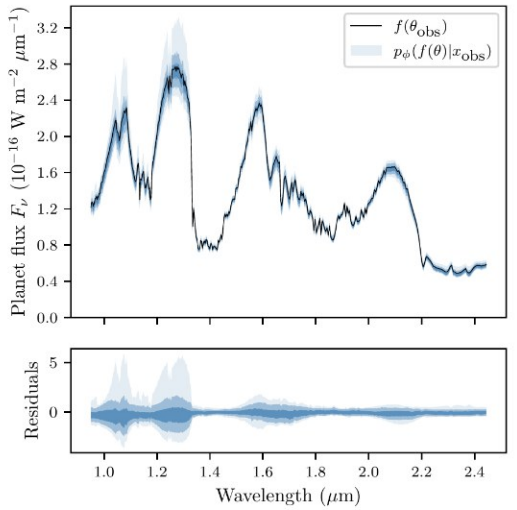
Detecting and handling misspecification: Schmitt et al, 2023; Ward et al, NeurIPS 2022.

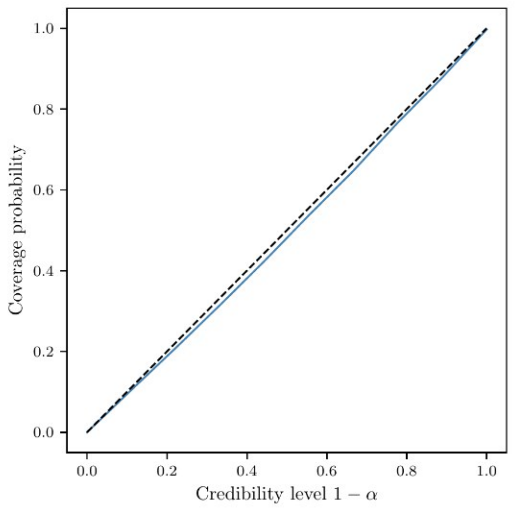


Neural posterior estimation for exoplanetary atmospheric retrieval

Malavika Vasist[1,2,⋆], François Rozet[2,⋆⋆], Olivier Absil[1,⋆⋆⋆], Paul Mollière[3], Evert Nasedkin[3], and Gilles Louppe[2]





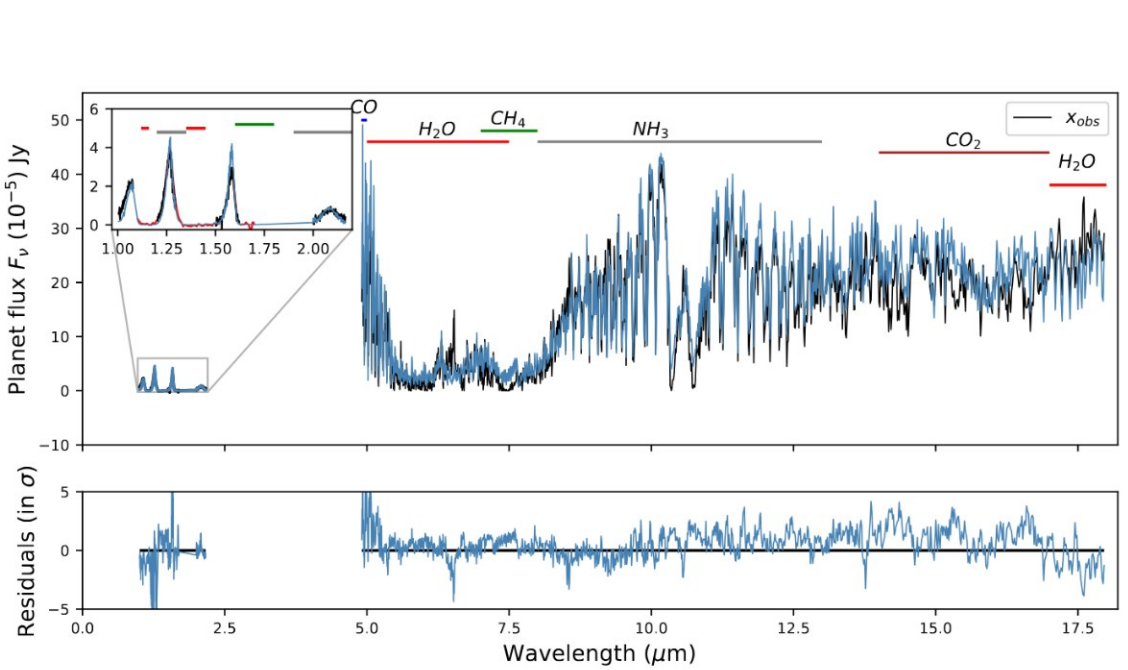


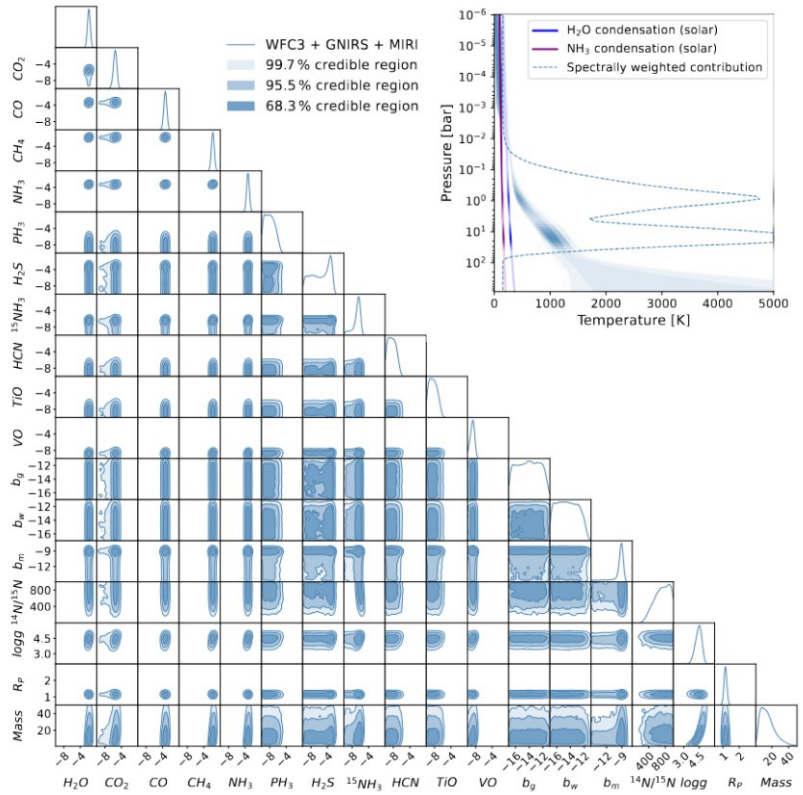
Vasist et al, A&A 672, A147, 2023. Simulated benchmark spectrum.



Panchromatic characterization of the Y0 brown dwarf WISEP J173835.52+273258.9 using JWST/MIRI

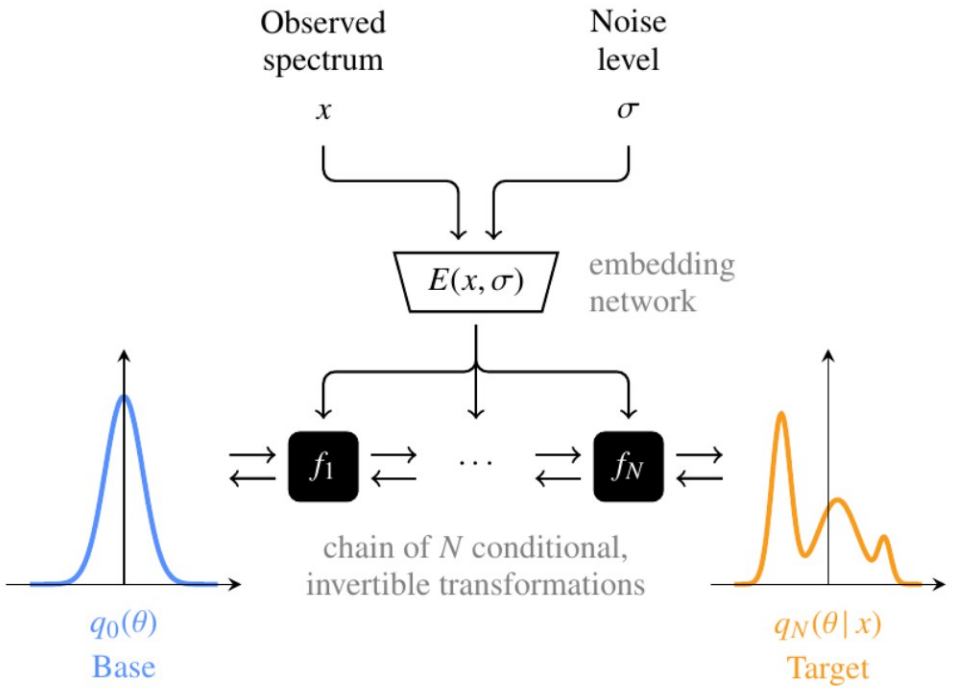
M. Vasist[1,2], P. Mollière[3], H. Kühnle[4], O. Absil[1], G. Louppe[2], Rens Waters[5,6,7], Manuel Güdel[8,4], Thomas Henning[3] David Barrado[9], Leen Decin[10], John Pye[11], and Pascal Tremblin [12]



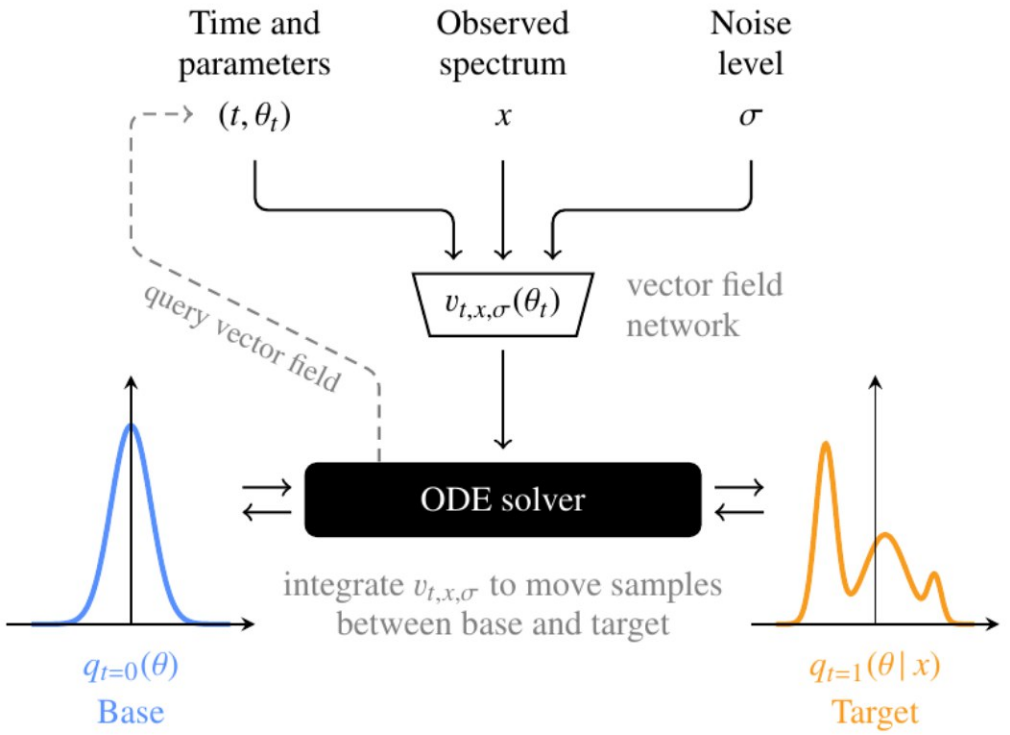


Beyond normalizing flows

NPE is agnostic to the choice of density estimator $q_\phi(\theta|x)$. A discrete flow composes a few invertible transformations, whose architecture must keep the Jacobian tractable. A continuous flow integrates a vector field given by an unconstrained network, which makes it more flexible.



Discrete flow (NPE)

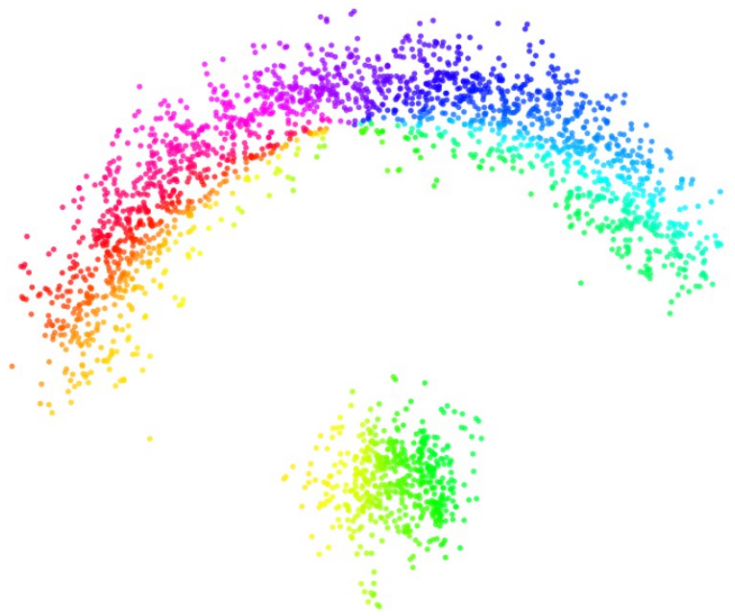


Continuous flow (FMPE)

Credits: Gebhard et al, A&A 2025.

Flow matching posterior estimation (FMPE)

$t = 0$ ——— $t = 1$



Noise $\theta_0 \sim \mathcal{N}(0, I)$ flows along a vector field $v_\phi(\theta_t, t; x)$, from $t = 0$ to $t = 1$, where it follows the posterior $q_\phi(\theta|x)$.

Animation: the exact flow-matching vector field towards a two-mode target.

As for discrete flows, the density is exact,

$$\begin{aligned}\log q_\phi(\theta|x) &= \log \mathcal{N}(\theta_0; 0, I) + \int_1^0 \operatorname{tr}\left(\frac{\partial v_\phi(\theta_t, t; x)}{\partial \theta_t}\right) dt \\ &= \log \mathcal{N}(\theta_0; 0, I) + \int_1^0 \nabla \cdot v_\phi(\theta_t, t; x)\, dt,\end{aligned}$$

where θ_0 is the noise that θ flows back to at $t = 0$. The vector field is trained by regression along straight paths $\theta_t = (1 - t)\,\theta_0 + t\,\theta$,

$$\min_\phi \; \mathbb{E}_{t\sim\mathcal{U}(0,1),\; \theta_0\sim\mathcal{N}(0,I),\; (\theta,x)\sim p(\theta,x)} \left[\left\| v_\phi(\theta_t, t; x) - (\theta - \theta_0)\right\|^2\right],$$

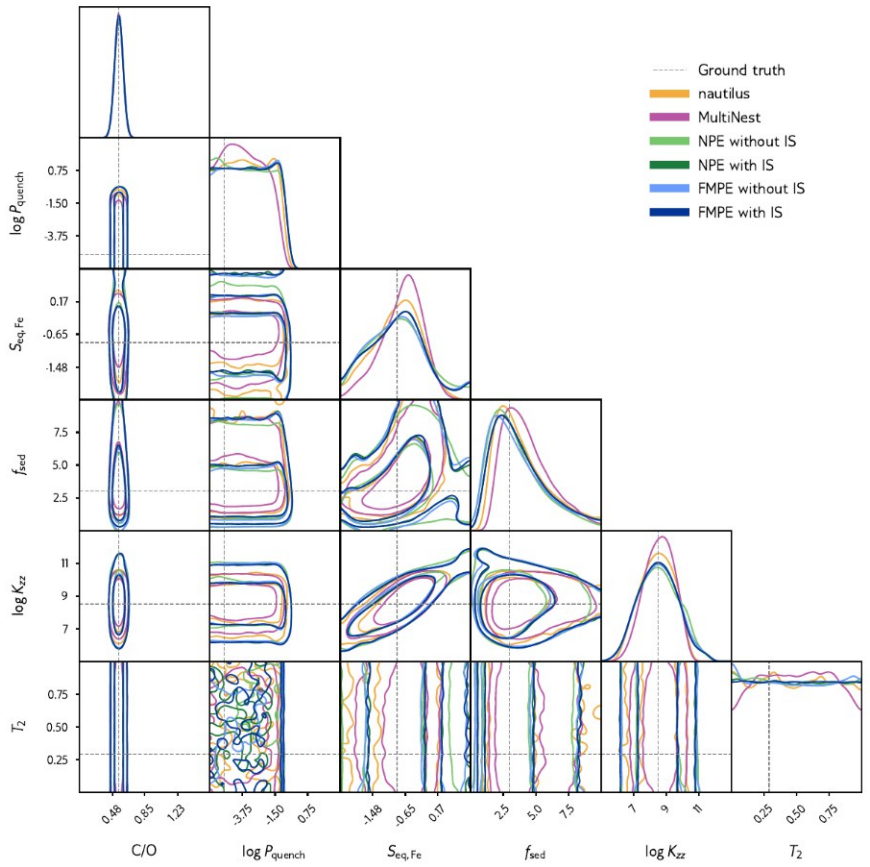
which requires no integration of the flow.

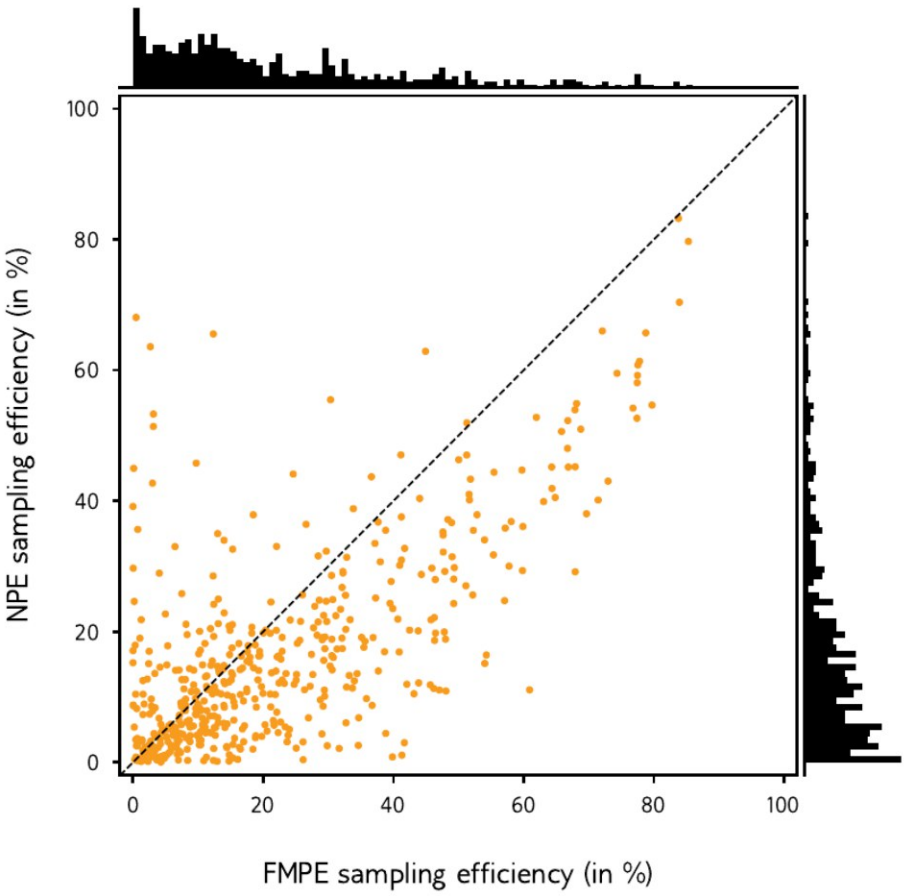
Dax et al, NeurIPS 2023.



Flow Matching for Atmospheric Retrieval of Exoplanets: Where Reliability meets Adaptive Noise Levels

Timothy D. Gebhard[1,2,*], Jonas Wildberger[1,3], Maximilian Dax[1], Annalena Kofler[1,4], Daniel Angerhausen[2], Sascha P. Quanz[2,5], and Bernhard Schölkopf[1,6]



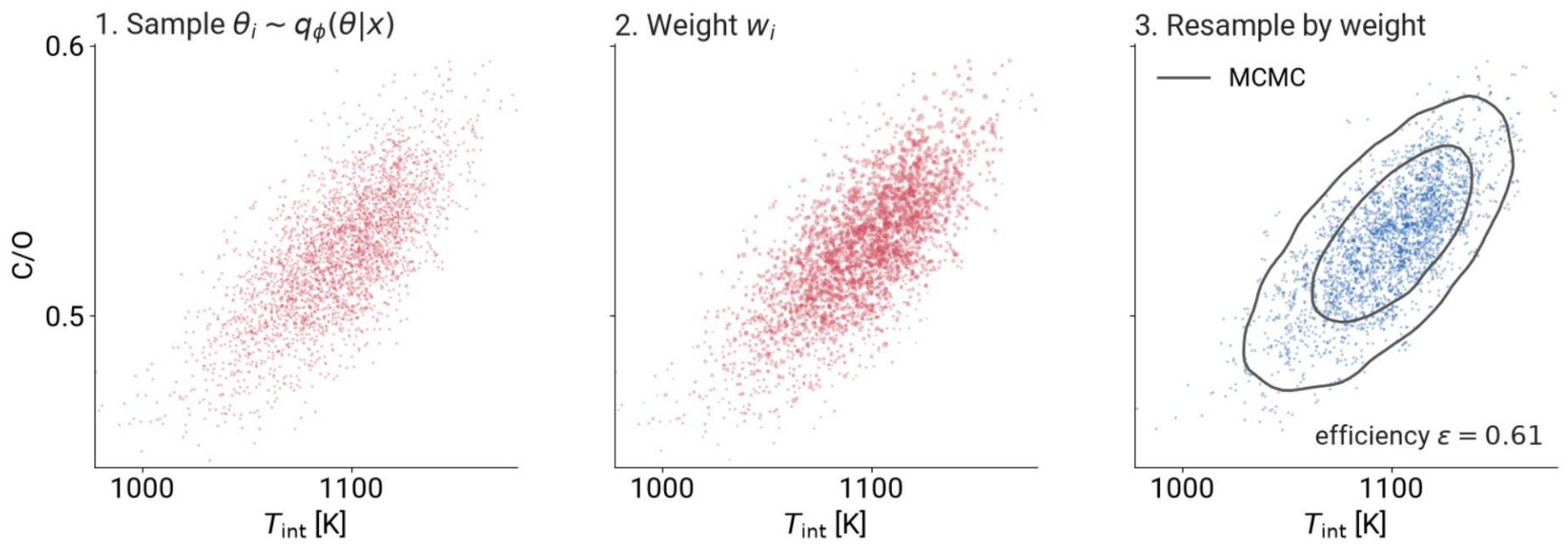


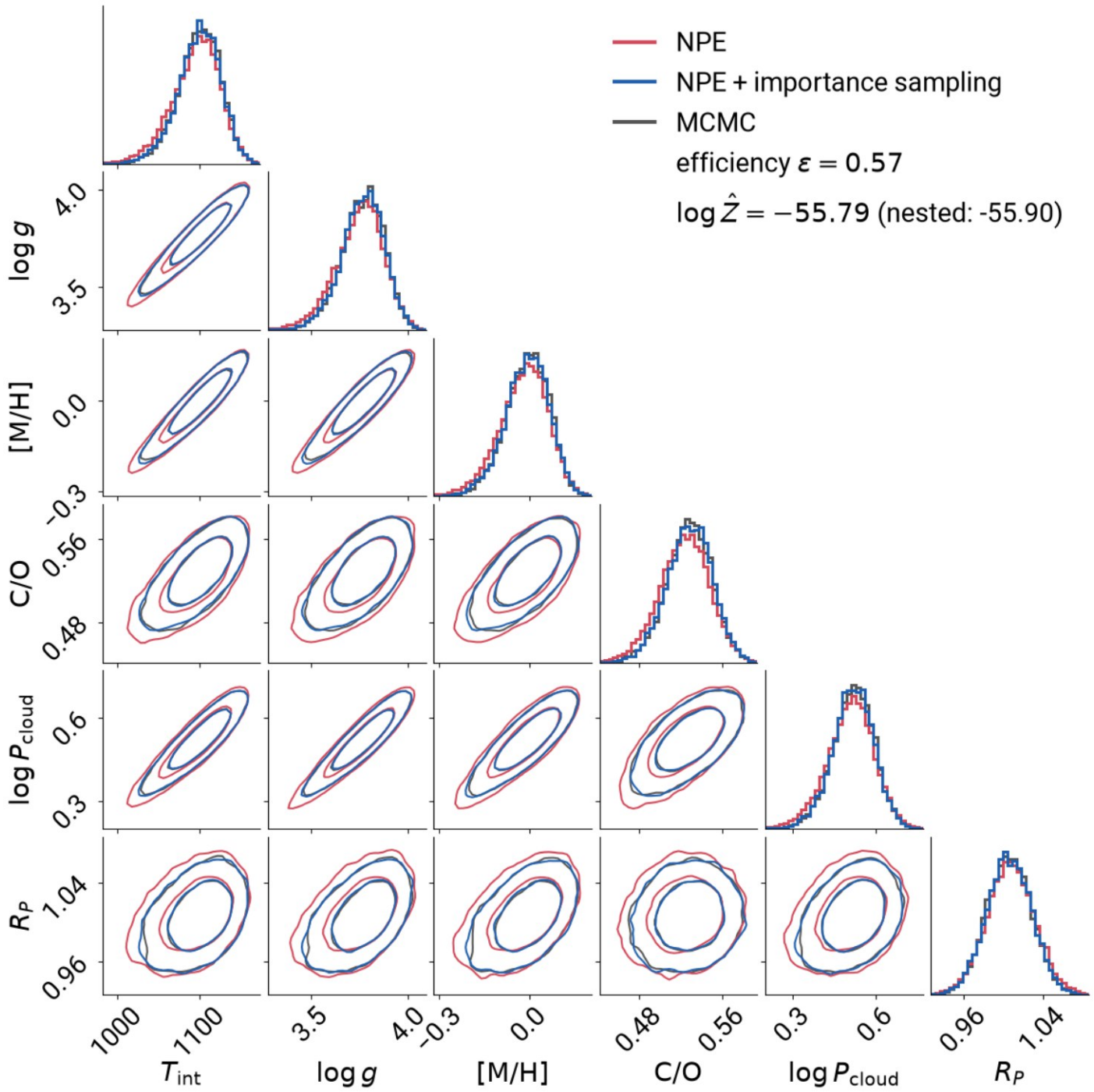
Cheat code: importance sampling

When the likelihood can be evaluated, the network becomes a **proposal**. Samples $\theta_i \sim q_\phi(\theta|x)$, weighted by

$$w_i = \frac{p(x|\theta_i)\, p(\theta_i)}{q_\phi(\theta_i|x)},$$

follow the **exact** posterior.

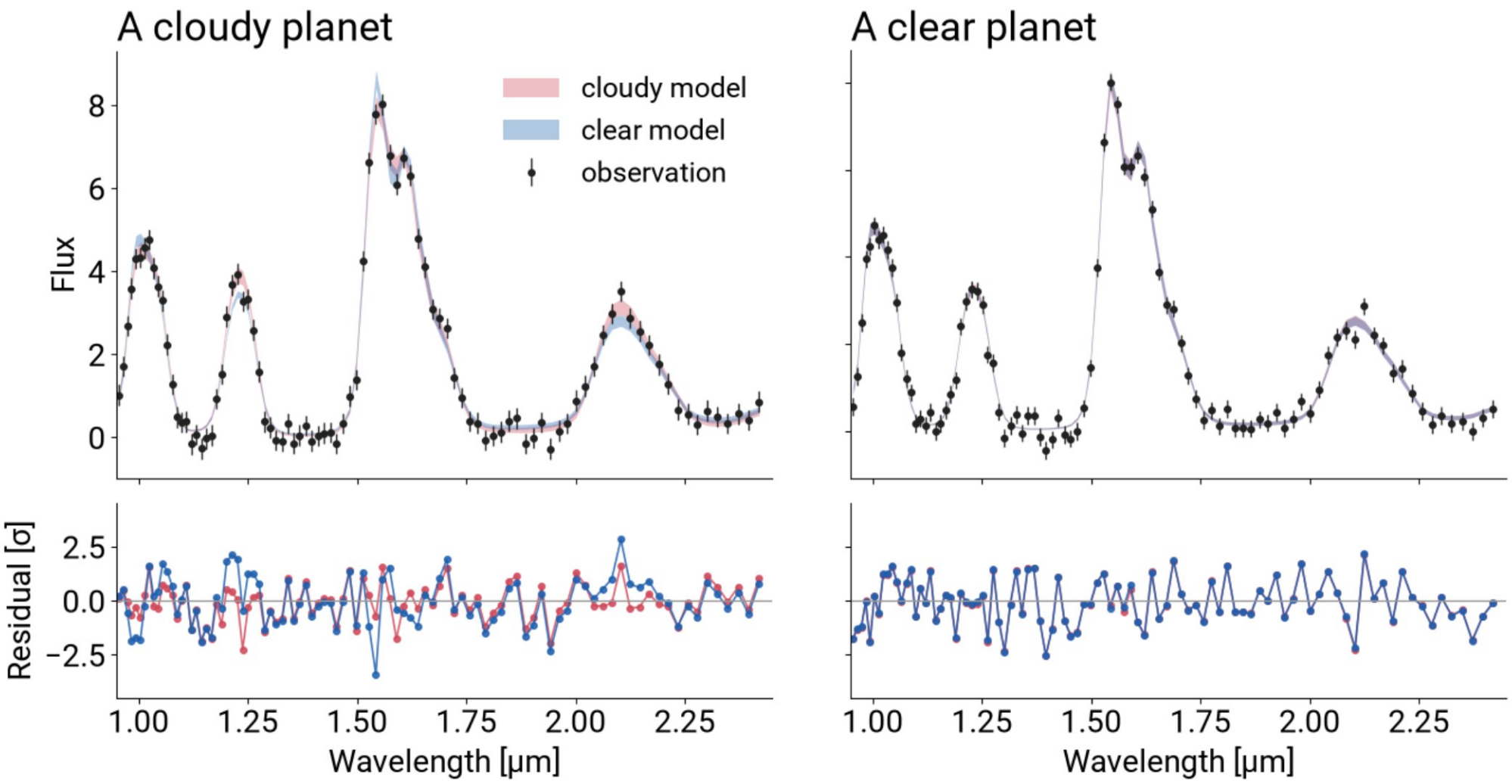


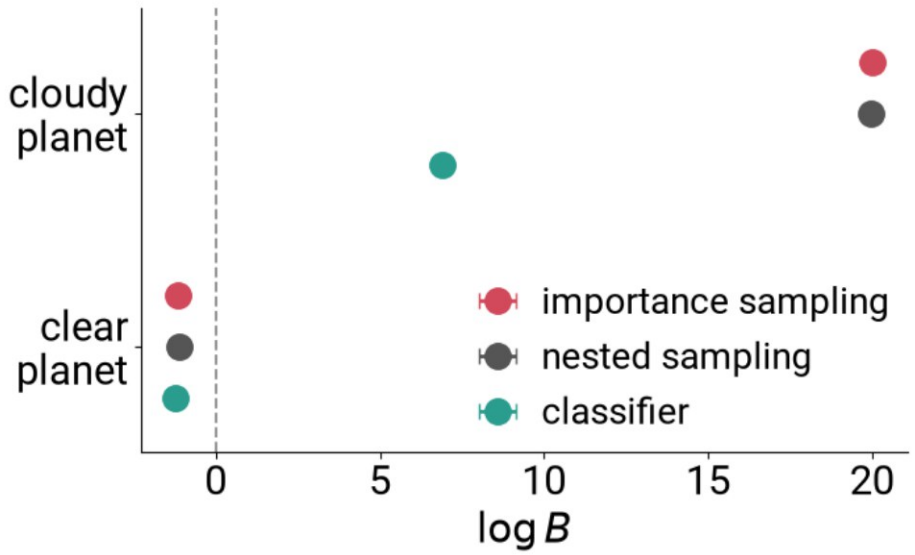


With importance sampling (blue), NPE matches MCMC (grey).

Model comparison

We compare two simulators, with and without a cloud deck. By eye, both fit.





The **Bayes factor** $B = p(x|\text{cloudy}) \,/\, p(x|\text{clear})$ compares the evidence of the two models.

Nested sampling $$p(x|M) \approx \sum_i p(x|\theta_i, M)\, \Delta X_i$$

Importance sampling $$p(x|M) \approx \frac{1}{n} \sum_i \frac{p(x|\theta_i, M)\, p(\theta_i|M)}{q_\phi(\theta_i|x)} = \frac{1}{n} \sum_i w_i$$

Classifier $$\log B \approx \log \frac{d_\psi(x)}{1 - d_\psi(x)}, \quad d_\psi(x) \approx p(\text{cloudy}|x)$$

Classifier: Radev et al, 2020. Model comparison without the evidence: MIRA, Sharief et al, ICML 2026.

Summary

Inference requires a simulator, not a likelihood.

A neural posterior estimator is trained once; it then gives the posterior of any spectrum in seconds.

Calibration tests the network; posterior predictive checks test the model.

An overdispersed posterior is safer than an overconfident one.

With a likelihood, importance sampling corrects the network and gives the evidence, for model comparison.

Reading list

Reviews

- Dax et al, An introduction to Bayesian and frequentist simulation-based inference with machine learning, 2026.
- Cranmer, Brehmer and Louppe, The frontier of simulation-based inference, PNAS 2020.
- Deistler et al, Simulation-based inference: a practical guide, 2025.
- Arruda et al, Diffusion models in simulation-based inference: a tutorial review, TMLR 2026.

Simulation-based inference for exoplanets

- Márquez-Neila et al, Supervised machine learning for analysing spectra [...], Nature Astronomy 2018.
- Haldemann et al, Exoplanet characterization using conditional invertible neural networks, A&A 2023.
- Vasist et al, Neural posterior estimation for exoplanetary atmospheric retrieval, A&A 2023.
- Aubin et al, Simulation-based inference for exoplanet atmospheric retrieval [...], 2023.
- Ardévol Martínez et al, FlopPITy: enabling self-consistent exoplanet atmospheric retrievals [...], A&A 2024.
- Gebhard et al, Flow matching for atmospheric retrieval of exoplanets, A&A 2025.
- Lueber et al, Near-instantaneous atmospheric retrievals and model comparison with FASTER, ApJL 2025.
- Vasist et al, Panchromatic characterization of the Y0 brown dwarf WISE 1738 using JWST/MIRI, A&A 2025.



The end.